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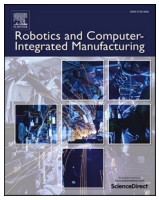

Full length Article

A safety posture field framework for mobile manipulators based on human–robot interaction trend and platform-arm coupling motion

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ABSTRACT

Mobile manipulators are increasingly deployed in industrial settings, such as material handling and workpiece loading, where they must safely interact with humans while efficiently completing tasks. Existing motion planning methods for mobile manipulators often struggle to ensure both safety and efficiency in dynamic human-robot interaction environments. This paper proposes a Safety Posture Field framework that addresses these limitations by firstly predicting human motion trends using the improved Long Short-Term Memory neural network and applying these predictions to potential field calculations for both the mobile platform and the robotic arm. During different stages of human-robot interaction, the mobile manipulator places varying emphasis on safety and efficiency while in motion. Additionally, when the robotic arm executes operations, a platform-arm coupling motion strategy is introduced when the potential field detects risks of singularity or local optima, preventing the robotic arm from becoming unstable or failing to reach the target pose in time. This strategy enhances the system's flexibility and operational stability. Comparative experiments in simulation and real-world settings confirm the ability of the framework to maintain high safety standards while improving task efficiency, making it suitable for industrial Human-Robot Interaction applications.

1. Introduction

With the rapid development of modern industry, industrial automation systems play a pivotal role in manufacturing, logistics, and warehousing sectors. For instance, tasks such as tool handling, workpiece loading, and automated inspection are often complex and highly dynamic in these industrial environments, requiring automation devices not only to execute tasks efficiently but also to collaborate safely with human operators [1,2]. Particularly in human-robot interactive (HRI) environments, automation devices need to frequently move between various workstations while being constrained by human-robot interaction [3]. This necessitates a high degree of freedom to adjust their trajectories in real-time and respond to uncertainties in dynamic environments, ensuring the safety of humans and other objects while successfully completing their tasks [4,5].

To address these demands, mobile manipulators have emerged as one of the optimal solutions for industrial automation [6–8]. By integrating mobile platforms and robotic arms, mobile manipulators can move flexibly between different workstations to execute tasks efficiently. The coordinated movement between the platform and the arm allows mobile manipulators to fully leverage their redundant motion capabilities, effectively handling various tasks in complex environments. At the same time, mobile manipulators must ensure both task efficiency and safety in human-robot collaborative environments. Through precise motion planning and dynamic obstacle avoidance, mobile manipulators can avoid potential threats to workers [9,10]. However, despite their potential, mobile manipulators face significant challenges in achieving motion planning that ensures both higher safety and efficiency in industrial settings.

In addressing the need for dynamic motion planning and safe obstacle avoidance in industrial environments, various path planning and obstacle avoidance algorithms have been employed [11–13], but they still exhibit notable limitations. The main categories include graph search-based algorithms [14,15], sampling-based algorithms [16,17],

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optimization-based algorithms [18,19], and learning-based algorithms [20–22].

Graph-based algorithms, such as A* [13] and Dijkstra [23], primarily search for the shortest path by discretizing the workspace and conducting a global search within the predefined graph structure. These algorithms are characterized by their ability to efficiently find the optimal path in static environments. Graph-based algorithms offer high-quality path planning and are easy to understand and implement. However, their computational complexity increases significantly, especially in high-dimensional spaces where mobile manipulators operate, resulting in a rapid increase in computation time. Additionally, these algorithms are less effective at dynamic updates. When encountering dynamic obstacles, graph-based algorithms often require reconstructing the graph or recalculating the path, leading to slow response times and an inability to promptly handle dynamic changes.

Sampling-based algorithms, such as Rapidly-Exploring Random Tree (RRT) [19], search for paths in continuous spaces by randomly sampling the environment. These algorithms progressively expand a tree structure, exploring feasible paths from the starting point to the target, avoiding the complexity of pre-constructing a grid in continuous spaces. They are known for their fast convergence and are well-suited for large-scale and high-dimensional space searches. However, they still face challenges in efficiently replanning paths when dealing with dynamic obstacles, despite the introduction of improved algorithms that account for dynamic characteristics. Nonetheless, they still encounter difficulties when computing paths in six-dimensional or higher-dimensional spaces, such as high computational burdens and lower real-time performance.

Optimization-based algorithms, such as Model Predictive Control (MPC) [24,25] and Nonlinear Programming (NLP) [26–28], generate smooth paths that ensure task efficiency and precision. However, optimization algorithms often encounter computational bottlenecks when handling complex environments. As the model complexity increases, the computation time for MPC may significantly rise; additionally, the selection of the prediction time window can affect both the quality and computational cost of the path planning. Furthermore, the solver's efficiency, particularly in high-dimensional spaces, often limits the real-time performance and safety of path planning [29,30]. In industrial scenarios, the system may fail to compute a safe avoidance path in time, jeopardizing task execution and reducing safety [31].

While learning-based algorithms have shown promising results in some experimental settings, their safety and stability in industrial applications remain challenging. Although learning algorithms can effectively capture complex task patterns, they require large amounts of training data, and the data can vary significantly across different industrial scenarios. This not only increases the difficulty and cost of data collection but also limits the generalizability of models across different environments. In complex industrial environments, collecting sufficient data is often challenging, and even when data is available, ensuring that the model performs safely and stably in new environments is a significant hurdle [27,32]. Moreover, learning algorithms demand significant computational resources, which hinders their ability to ensure real-time performance and safety in dynamic human-robot interactions, which is critical in industrial environments [33,34].

Compared to the methods mentioned above, the Artificial Potential Field (APF) method excels in handling dynamic obstacle avoidance due to its low computational complexity and high real-time performance [35,36]. By constructing virtual attractive and repulsive fields, the potential field method can adjust paths quickly, ensuring both safety and task efficiency in dynamic environments.

However, the potential field method relies on sensor data to plan motion, which only reflects the current state of the environment. In dynamic environments where workers and obstacles are constantly moving, the data captured by sensors is often momentary and does not predict future changes [37,38]. This leads to lag in response, meaning that the robot may not adjust its path quickly enough as the environment evolves, resulting in decreased obstacle avoidance performance and increased safety risks [39]. Furthermore, when too many factors influence the potential field, especially in HRI environments where the target is close to personnel, the system may struggle to compute effective solutions, resulting in failure to reach the target, overly complex paths, or low operational efficiency. Traditional potential field methods also fail to fully utilize the redundant motion capabilities of mobile platforms, particularly in tasks involving the robotic arm. The method struggles to handle the flexibility and redundant degrees of freedom of the robotic arm, limiting the safety and task flexibility of task execution in complex environments [40,41].

To address these issues, this paper proposes three key improvements based on the original potential field: human motion prediction, an adaptive adjustment of the potential field for dynamic obstacles and robot singularity, and a coupled motion strategy for mobile platforms and robotic arms.

In terms of human motion prediction, human movements in industrial environments often exhibit temporal correlation, and the actions of workers at specific workstations tend to follow regular patterns [42]. This allows for more accurate predictions of human actions based on environmental cues. Long Short-Term Memory (LSTM) networks are particularly well-suited for handling sequential data and can effectively capture the temporal dependencies of human movements. The strength of LSTM networks lies in their ability to retain long-term historical motion data and predict future movement trends, which is especially useful in industrial settings where actions are regular and repetitive [43,44]. Additionally, the incorporation of Attention mechanisms and pruning layers can help retain the most significant input dimensions based on the semantic context of the environment, thereby improving efficiency during the prediction phase. Therefore, LSTM networks are well-suited for human motion prediction in such environments.

As for the calculation of the artificial potential field, traditional methods often struggle to generate effective paths in the presence of dynamic obstacles during the robot's approach to the target, or they exhibit low operational efficiency in such cases [45]. For instance, when dynamic obstacles and the robot move along non-intersecting paths, the robot does not need to perform avoidance maneuvers, which can enhance operational efficiency. Additionally, when dynamic obstacles are near the target point, the robot should not remain stationary but should approach the target at a safe speed. When the robotic arm is affected by obstacles and moves in a direction toward singularity, additional constraints must be introduced to prevent the arm from entering an unstable state.

Regarding the coupled motion strategy, when mobile platforms reach a workstation to perform tasks, relying solely on motion planning for the robotic arm in joint space often leads to the robotic arm approaching singular positions or exceeding joint motion limits due to the influence of obstacles. In such cases, exploring a coupled motion strategy becomes crucial. When the robotic arm encounters difficulties in solving for the target motion, the coupled motion strategy coordinates the movements of both the mobile platform and the robotic arm to improve task execution efficiency.

Thus, this paper proposes a Safety Posture Field (SPF) framework, based on the potential field method, which addresses the limitations of traditional potential field methods by combining human motion prediction, utilizing an improved APF calculation, with a coupled motion strategy. This framework enhances the robot's adaptability and stability in obstacle avoidance while ensuring safety and task efficiency in human-robot collaboration settings.

The main contributions of this paper are as follows:

1. Human action prediction: Using the improved LSTM neural networks, human movements are predicted based on previous action data to estimate the future motion coordinates of key human body parts over a specific time period, guiding real-time dynamic obstacle avoidance and ensuring safe approaches by the mobile manipulator,

while pruning based on environmental information is employed to enhance prediction efficiency.
2. Potential field aimed at safety and efficiency: The potential field values are calculated based on the predicted motion trends of dynamic obstacles, allowing the robot to move at a smooth speed while enabling movement near obstacles at speeds that comply with safety standards. A singularity potential field is also introduced to prevent the robotic arm from becoming unstable.
3. Platform-arm coupled motion strategy: It couples the joint movements of the robotic arm with the planar movement of the mobile platform. When the framework detects that the arm is approaching a singular configuration under the influence of the potential field, or when the motion steps become too long, the platform performs directional movement. This ensures that during joint space operations, the robotic arm avoids singular configurations and simplifies its motion trajectory to improve efficiency.

The rest of this paper is organized as follows: Section 2 provides an overview of the HRI relationships addressed in this paper and the relationships between the modules of the SPF framework. Section 3 to Section 5 detail the specific principles of the three modules. Section 6 covers simulations, experiments, and applications about the framework. Section 7 discusses the strengths and limitations of this study. Section 8 provides a summary and outlook.

2. HRI relationships and SPF framework modules

2.1. HRI relationships

Our proposed framework operates in HRI environments characterized by dynamic obstacles, such as human workers, alongside stationary obstacles, such as workstations, as shown in Fig. 1. It is important to note that the study is limited to scenarios where robots perform independent tasks in shared environments with humans but does not involve physical human-robot collaboration. Therefore, the main focus of this study is on collision avoidance and efficient task execution, rather than shared manipulation.

In such human-robot collaboration environments, the primary elements to consider include the positions of humans $[p_m, p_{hl}, p_{hr}]$, considered as dynamic obstacles, the locations of fixed obstacles p_{obs} such as workstations, and the robot's target point $p_{b,goal}$ and pose $p_{a,goal}$, etc. For simplicity, the key notations used to represent these factors and their relationships are summarized in Table 1.

Table 1
List of some key notations.

Notation	Description
p_b	Mobile platform position
v_b	Mobile platform velocity
p_a	End-effector position of robotic arm (Task space)
q_a	End-effector position of robotic arm (Joint space)
v_a	End-effector velocity of robotic arm (Task space)
$\dot{q}_a$	End-effector velocity of robotic arm (Joint space)
p_m	Current position of human center
v_m	Current velocity of human center
$\hat{p}_m$	Predicted position of human center
$\hat{v}_m$	Predicted velocity of human center
p_{hl}	Current position of human left hand
v_{hl}	Current velocity of human left hand
$\hat{p}_{hl}$	Predicted position of human left hand
$\hat{v}_{hl}$	Predicted velocity of human left hand
p_{hr}	Current position of human right hand
v_{hr}	Current velocity of human right hand
$\hat{p}_{hr}$	Predicted position of human right hand
$\hat{v}_{hr}$	Predicted velocity of human right hand
$p_{b,start}$	Starting position of mobile platform
$p_{b,goal}$	Target position of mobile platform
p_{obs}	Position of fixed obstacles
$p_{a,start}$	End-effector starting position of robotic arm (Task space)
$q_{a,start}$	End-effector starting position of robotic arm (Joint space)
$p_{a,goal}$	End-effector target position of robotic arm (Task space)
$q_{a,goal}$	End-effector target position of robotic arm (Joint space)

2.2. SPF framework modules

The framework of Safety Posture Field, as illustrated in Fig. 2, comprises ***HRI State Monitoring*** module, ***Potential Field Calculation*** module and ***Coupled Motion*** module. These modules work together to ensure real-time monitoring, calculation, and action adjustment within HRI scenarios to enhance safety and task execution efficiency.

Firstly, the ***HRI State Monitoring*** module plays a crucial role in gathering real-time data about human movements and analyzing their interaction with the robot's tasks. The purpose of this module is to monitor obstacle information in the robot's operating environment, including static obstacles p_{obs} and dynamic human behavior $[p_m, v_m, p_{hl}, v_{hl}, p_{hr}, v_{hr}]$, and to generate predictions about future interactions within the collaborative environment. The module focuses on analyzing the temporal sequences $\hat{P}_m, \hat{P}_{hl}, \hat{P}_{hr}$ of human actions to assess the safety posture of the HRI environment. The results of these predictions are divided into two categories based on the robot's task phase: During the

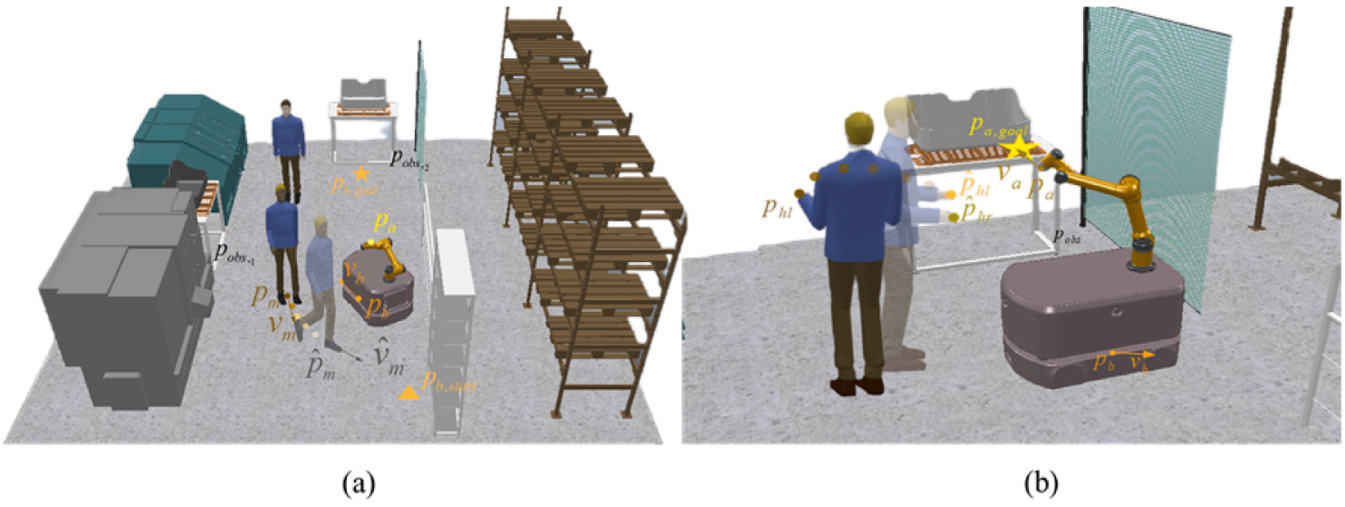


Fig. 1. Overview of HRI environments: (a) the mobile platform transfers betw-een workstations and (b) the robotic arm executes close-proximity operations with humans at the workstation.

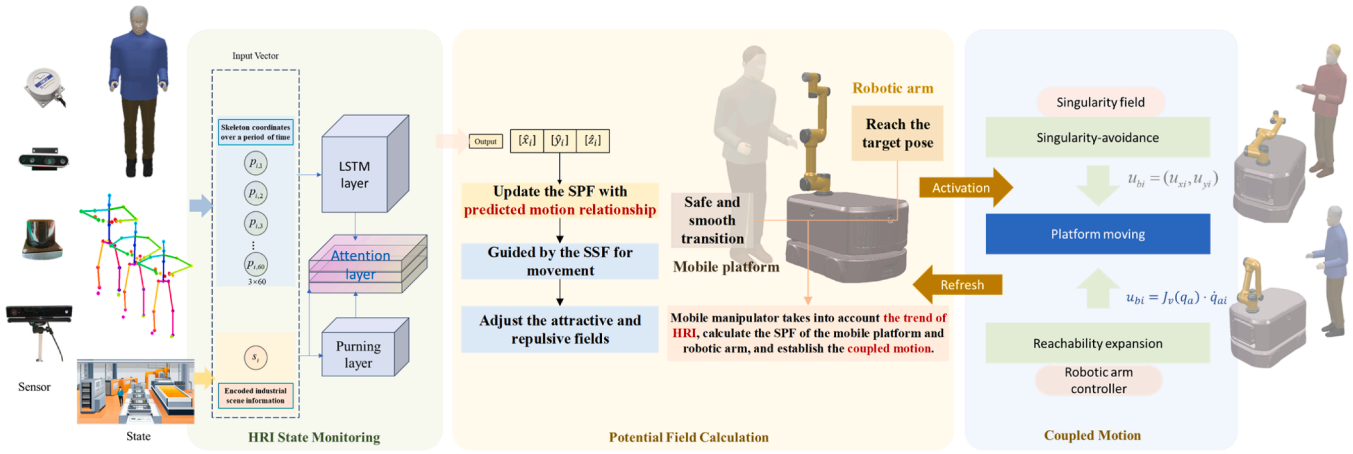


Fig. 2. The framework of Safety Posture Field.

phase when the mobile platform approaches a target workstation, the system outputs predictions related to the movement of the human's skeleton center point $\widehat{P}_m$. When the manipulator is about to perform a task, the system outputs predictions related to the human's hand movements $\widehat{P}_{hl}, \widehat{P}_{hr}$. These predictions are then sent to the Potential Field Calculation module for further processing, enabling the robot to adjust its motion accordingly.

Additionally, the ***Potential Field Calculation*** module ensures human safety and robot motion by calculating potential fields F_b, F_a based on relative velocity and distance in task space, such as the velocity between the mobile platform and the predicted human motion $(v_b, \widehat{v}_m)$ and the distance between the end-effector of the robotic arm and the target pose. The module handles separate potential field calculations for the mobile base and the manipulator, focusing on obstacle avoidance and smooth task execution. To prevent the robotic arm from encountering singularities, a singularity potential field F_s is introduced. However, this may cause excessive trajectory changes, reducing efficiency. Additionally, there is a risk that the joint movements of the robotic arm may exceed the specified limits. To address these challenges, the ***Coupled Motion*** module is proposed, where the mobile base and manipulator coordinate their movements to avoid singularities while maintaining task efficiency. This module calculates the required adjustments in both the base and the manipulator's positions to achieve safe and efficient task execution. Key parameters involved in this module include the velocities v_a and v_b of the platform and arm, the distances between obstacles and the robot $(p_a, p_{obs}), (p_b, p_{obs})$, and the safety thresholds τ_s that trigger adjustments in motion.

In the following chapters, each module will be discussed in detail, covering the principles behind their design, the parameters used, and their specific roles in ensuring safe and efficient human-robot interaction in dynamic industrial environments.

3. HRI state monitoring module

3.1. Object state monitoring

In HRI industrial environments, robots need to continuously monitor the position, movements, and working status of human operators to ensure their actions do not pose any threat to humans. The HRI State Monitoring module is a fundamental component of the entire system, collecting environmental information through sensor data and utilizing advanced predictive models to analyze human motion trends in real-time. This module not only processes real-time monitoring data but also predicts future human actions, enhancing the robot's safety and task execution efficiency in dynamic environments.

The system continuously acquires key object state information in the environment through various sensors. A two-dimensional laser scanner mounted on the mobile platform scans static obstacles in the plane, generating time-stamped coordinate data, represented as $p_{obs}(x,y)$. For dynamic human movements and fixed three-dimensional obstacles, RGB-D cameras independently positioned in the environment are used for monitoring, providing time-stamped three-dimensional coordinate data, represented as: $p_m, p_{hl}, p_{hr}, p_{obs}(x,y,z)$. Additionally, scene features are recognized using the robot's onboard RGB camera, which identifies pre-encoded scene information through image recognition technology, mapping it to corresponding symbolic representations for subsequent decision-making and planning.

3.2. Improved LSTM model

Since human movements in industrial environments are governed by the tasks they perform, and these tasks are highly related to the workstation, the scene information of the workstation provides important reference value for the accurate prediction of human actions. To this end, the improved LSTM-based action prediction model incorporates scene information as one of the inputs and uses it as a guiding factor for the time-dependent prediction. The model dynamically adjusts its focus on different time steps using the Attention mechanism, optimizing the prediction of action probabilities based on historical data. The system employs a partially observable Markov decision process to generate optimal sequential action predictions.

To facilitate model input, industrial scene information is encoded for ease of input (e.g., s_1 for the assembly area, s_2 for the material area, and s_3 for the turnover area), allowing the robot system to identify this information in real time. Human action data is captured using Microsoft Kinect V2, which tracks 25 skeleton feature points. Based on common actions in industrial scenarios, these points are simplified to 11 key feature points, as illustrated in Fig. 3(a). These feature points cover the main parts of the human body involved in motion and can effectively capture the core action characteristics related to tasks in industrial environments.

The model input $\boldsymbol{x}_t$ consists of 2 s (60 frames) of skeleton data $\boldsymbol{P}_i$ (3D coordinates (x,y,z) of 11 feature points) and the encoded scene information s_i, where i is the subscript used to distinguish different objects, such as m representing the human center, hl representing the human left hand, and the scene number as mentioned earlier. The output of the model is the predicted coordinate sequence $\widehat{\boldsymbol{p}}_i$ of the specified target for the next 1 s (30 frames). The LSTM layer consists of two layers, with each layer containing 128 hidden units, captures the temporal dependencies in the input sequence by processing the input $\boldsymbol{x}_t$ and generating a hidden state $\mathbf{h}_t$ at each time step, as illustrated in Fig. 3(b). The LSTM updates its hidden states and cell states using the following recursive equation:

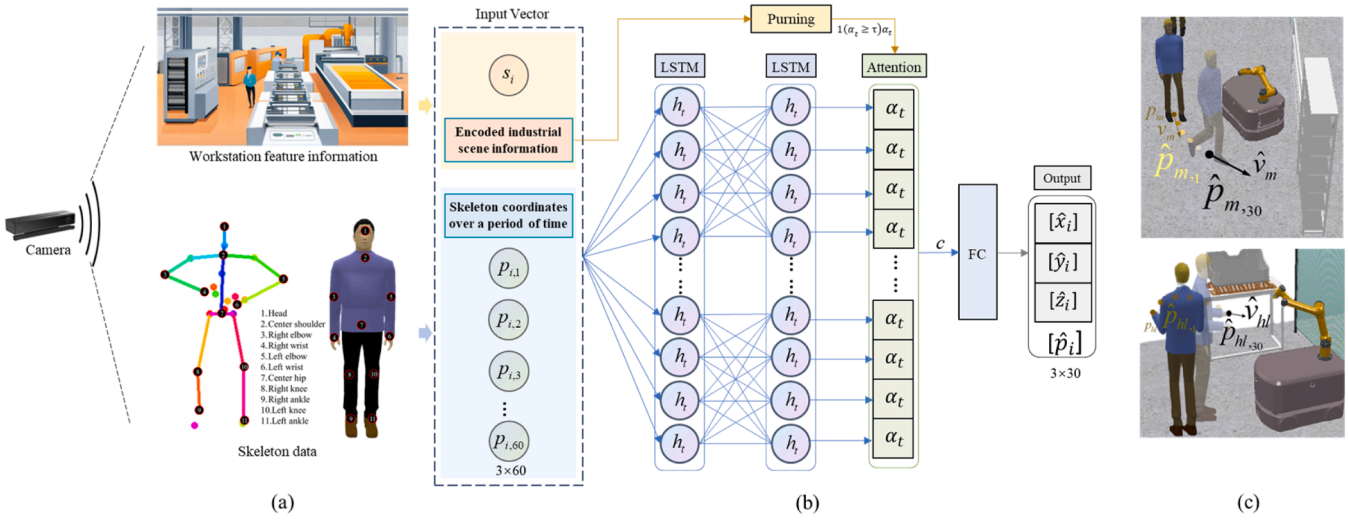


Fig. 3. The framework of HRI State Monitoring Module.

$$\boldsymbol{h}_t, \boldsymbol{c}_t = LSTM(\boldsymbol{x}_t, \boldsymbol{h}_{t-1}, \boldsymbol{c}_{t-1}) \tag{1}$$

The hidden states $\mathbf{h}_t$ then used by the Attention mechanism. This mechanism dynamically adjusts the model's focus on different time steps, with attention weights α_t computed as follows:

$$e_t = \boldsymbol{v}^T \tanh(\boldsymbol{W}_h \boldsymbol{h}_t + b_h) \tag{2}$$

$$\alpha_t = \frac{exp\ (e_t)}{\sum_{t=1}^{T} exp\ (e_t)} \tag{3}$$

here, $\mathbf{h}_t$ represents the hidden state at time step t, $\mathbf{W}_h$ is the weight matrix applied to the hidden state, and $\mathbf{v}^T$ is a learnable weight vector used to compute the attention score e_t.

The attention weights are used to compute a weighted sum of the LSTM hidden states, resulting in the context vector $\mathbf{c}$:

$$\boldsymbol{c} = \sum_{t=1}^{T} \alpha_t \boldsymbol{h}_t \tag{4}$$

To further optimize model efficiency and prevent overfitting, we introduce a pruning layer based on the attention weights. This pruning layer dynamically adjusts the neural network's structure by removing less important neurons or connections, particularly those with low attention scores, thereby reducing computational overhead and speeding up inference.

$$\boldsymbol{c_p} = \sum_{t=1}^{T} 1(\alpha_t \geq \tau)\alpha_t \boldsymbol{h}_t \tag{5}$$

The function $1(\alpha_t \geq \tau)$ is an indicator function, which takes the value of 1 when $\alpha_t \geq \tau$, and 0 otherwise. This function indicates that only the hidden states corresponding to attention weights greater than or equal to the threshold τ are retained. The pruning process helps to maintain the model's accuracy while significantly improving its performance efficiency in both time and space.

The pruned context vector, retaining only the attention weights with significant values, is then combined with the last hidden state $\mathbf{h}_T$ and passed to a fully connected layer for action prediction:

$$\widehat{\boldsymbol{p}}_i = \boldsymbol{W}_o \left[\boldsymbol{c_p}; \boldsymbol{h}_T\right] + b_o \tag{6}$$

where $\widehat{p}_i$ is the predicted object's action coordinate sequence for the next 1 second.

The Attention mechanism enables the model to focus on critical time steps, improving the temporal prediction accuracy of human actions. To prevent overfitting, a Dropout layer with a rate of 0.3 is added after the LSTM layer. The final output is produced by three linear regression units, which predict the 3D coordinates (x, y, z) of the target feature points for the next second (30 frames). The model is optimized using the Adam optimizer, with the ReduceLROnPlateau scheduler dynamically adjusting the learning rate. The mean squared error (MSE) loss function is used to evaluate the difference between predicted and actual coordinates.

4. Potential field calculation module

The artificial potential field designed in this paper is primarily composed of the attractive field force F_{att}, repulsive field force F_{rep}, and singularity field force F_{sig}:

$$F_{SPF} = F_{att} + F_{rep} + sF_{sig} \tag{7}$$

These forces ensure the stable operation of the system by calculating the output speed of the mobile platform and robotic arm while adhering to the safety standards outlined in ISO/TS 15,066, balancing both safety and efficiency. s is the activation factor that ensures F_{sig} is only activated for calculation when the robotic arm is in motion. The design of these three components and stability analysis will be detailed below.

4.1. Attractive field

The robot application scenario studied in this paper involves setting the target position near personnel, who are considered dynamic obstacles, to accomplish tasks such as helping humans place necessary tools on the workstation or completing pre-assembly tasks. Therefore, the design of the attractive field must take into account HRI safety.

To this end, we define a safety zone within the HRI workspace. According to the human-robot collaboration standards [46], the shared space is divided into four main areas based on the relative distance between humans and robots, which are the safety-rated monitored stop (SMS), hand guiding(HG), speed and separation monitoring(SSM), and power and force limiting (PFL), listed from farthest to closest. The radii of these areas are calculated based on the average stride length d_{ave} of adults aged 22 to 30, as shown in Fig. 4. The mobile platform of the mobile manipulator moves across the HG, SSM, and PFL areas, while the robotic arm primarily operates within the PFL area.

In the PFL area, mobile manipulators with collaborative properties can also move, but the maximum allowable speed $v_{b\ max1}$ for mobile platform and $v_{a\ max}$ for robotic arm are limited. It is calculated based on the human-robot equivalent mass M, the transient impact force F_{max} that the most likely body part to be impacted during different robotic tasks

can withstand, and the equivalent stiffness constant k of that body part, which can be referenced from the biomechanics lookup table and calculated using the following equation:

$$v_{b\ max1} = \frac{F_{max}}{\sqrt{kM}} \tag{8}$$

where the $v_{b\ max1}$ can be replaced with $v_{a\ max}$ when calculating the potential field for the robotic arm. The equivalent mass M is calculated by:

$$F_{a,att\ i}\left(q_{a\ i}, q_{a\ i,goal}\right) = \begin{cases} \frac{\dot{q}_{a\ i\ max}}{2}\cdot\left(1 - cos\left|\left|q_{a\ i} - q_{a\ i,goal}\right|\right|\right), & \left|\left|q_{a\ i} - q_{a\ i,goal}\right|\right| \le \frac{1}{2}\theta_{max}\dot{q}_{a\ i\ max}, \\ \left|\left|q_{a\ i} - q_{a\ i,goal}\right|\right| > \frac{1}{2}\theta_{max} \end{cases} \tag{13}$$

$$M = \left(\frac{1}{m_H} + \frac{1}{m_R}\right)^{-1} \tag{9}$$

where m_H and m_R are the effective masses of the human body and the robot, respectively. m_H is obtained from the reference table provided by biomechanics, while m_R is a function of the total mass M_R of the robot's motion component and load m_L:

$$m_R = \frac{M_R}{2} + m_L \tag{10}$$

To ensure the safety of human-robot collaboration and the stability of mobile platform motion, the attractive potential force in this area is represented by an *S*-shaped function, and this force is directly used as the velocity control for the platform, adjusting the speed as it approaches the target:

$$f_{a1}\left(p_b, p_{b,goal}\right) = \frac{2v_{b\ max1}}{d_1}\cdot\frac{\left|\left|p_b - p_{b,goal}\right|\right|}{1 + e^{-\beta_1\left(\left|\left|p_b - p_{b,goal}\right|\right| - d_1\right)}} \tag{11}$$

where β_1 is the smoothing factor for mobile platform. This not only ensures that when the relative distance between the robot and human decreases as the robot approaches the target position, the robot's speed can rapidly and smoothly decay, but also allows for gradual deceleration until it stops at the target position.

For the robotic arm, before performing motion planning, the task space targets $p_a, p_{a,goal}$ should be replaced with $q_a, q_{a,goal}$ using inverse kinematics. Since its motion occurs entirely within the PFL space, the distance parameter d_1 is not applicable. Therefore, it is only necessary to calculate the maximum allowable speed $v_{a\ max}$ for the robotic arm in the task space based on the aforementioned safety restrictions. Then, at each step of operation, the maximum velocity $\dot{q}_{a\ max}$ in the joint space is computed using the Jacobian matrix:

$$\dot{q}_{a\ max} = J(q)^{-1}\cdot v_{a\ max} \tag{12}$$

And the potential field force for the robotic arm is calculated as follows and is interpreted as the velocity of the *i*-th joint:

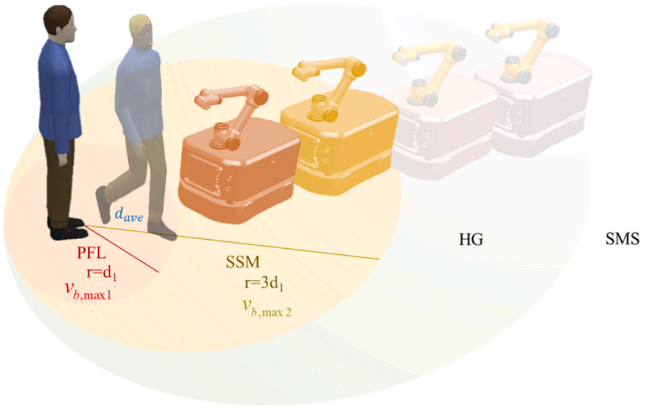


Fig. 4. Human-robot shared area division.

where α is the smoothing factor for robotic arm, i represents the i th joint of the robotic arm. In industrial scenarios, the motion range of collaborative robot joints, particularly joints 2, 3, and 4, which have the greatest impact on safety, mostly falls within the range of $1/2\ \theta_{max}$, where θ_{max} is the joint angle limit. For motions beyond this range, efficiency must be ensured while meeting safety standards. Therefore, the attractive force $F_{att\ i}$ is segmented at $1/2\ \theta_{max}$. The cosine-based form is adopted to ensure that the attractive force increases nonlinearly with distance, and the smooth decay of the potential field force $F_{a,att\ i}$ near the target helps to prevent overshooting or instability.

In the SSM area, the robot's maximum speed $v_{b\ max2}$ is not constrained by human biomechanics but is more focused on maintaining the stability of its motion system. For instance, based on the general limitations for mobile robots in industrial scenarios [42], we can set the maximum speed of the mobile platform to 0.8 m/s. When the mobile platform moves towards the target in the SSM area, we use a cubic polynomial to represent the potential field, ensuring smooth motion of the robot through second-order continuity of displacement. The corresponding virtual attractive force can be expressed as:

$$f_{a2}\left(p_b, p_{b,goal}\right) = -\left(\beta_2\cdot\left|\left|p_b - p_{b,goal}\right|\right| - m\right)^2 + n \tag{14}$$

where m, n and β_2 are coefficients that ensure the smoothness of the entire virtual attractive force function. By setting $(d_1, v_{b\ max1})$ and $(3d_1, v_{b\ max2})$, the values of m and n can be calculated.

In the HG and SMS areas, the robot does not need to respond to distant dynamic targets, and its speed can remain constant $v_{b\ max2}$. In summary, the virtual attractive force for the mobile platform can be expressed as:

$$F_{b,att}\left(p_b, p_{b,goal}\right) = \begin{cases} f_{a1}\left(p_b, p_{b,goal}\right), & \left|\left|p_b - p_{b,goal}\right|\right| \le d_1 \\ f_{a2}\left(p_b, p_{b,goal}\right), & d_1 < \left|\left|p_b - p_{b,goal}\right|\right| \le 3d_1 \\ v_{b\ max2}, & \left|\left|p_b - p_{b,goal}\right|\right| > 3d_1 \end{cases} \tag{15}$$

4.2. Repulsive field

The repulsive potential field is primarily caused by dynamic and fixed obstacles. In the industrial scenarios considered in this paper, the dynamic obstacles mainly consist of personnel in the environment. According to industrial safety standards for collaborative robots, we set the influence radius of personnel on the robot to the range of the SSM area. The relative distance is calculated based on the predicted positions of key human body parts after 1 second, obtained from the Improved LSTM Model, and tshe predicted speed calculated from these predicted positions, such as $\hat{p}_m$ and $\hat{v}_m$. The virtual repulsive force is derived from the negative gradient of the repulsive potential field, similar to the virtual

attractive force, and its result is added as velocity to the robot's motion. For the mobile platform, the virtual repulsive force from the moving personnel is:

$$F_{b,rep1}(p_b, v_b, \hat{p}_m, \hat{v}_m) = \begin{cases} -r_{b1}\left(\frac{1}{||p_b - \hat{p}_m||} - \frac{1}{3d_1}\right)\cdot\frac{|I_{vp}|}{||p_b - \hat{p}_m||^2},\ ||p_b - \hat{p}_m|| \le 3d_1 \ and\ I_{vp} < 0 \\ F_{b,rep2}(p_b, \hat{p}_m),\ \ ||p_b - \hat{p}_m|| \le 3d_1 \ and\ I_{vp} \ge 0 \\ 0,\ ||p_b - \hat{p}_m|| > 3d_1 \end{cases} \tag{16}$$

where r_{b1} is the repulsive force intensity factor, and I_{vp} is the dot product of the relative velocity and the relative position:

$$I_{vp} = (v_b - \hat{v}_m)\cdot(p_b - \hat{p}_m) \tag{17}$$

Considering that even when the robot enters the SSM range of the moving person, the influence of the moving person on the robot's motion depends on their relative velocity direction. The dynamic obstacle is only considered significant when the robot and the person are moving toward each other. Therefore, by calculating the dot product of the relative distance and relative velocity between the human and the robot, different intensities of repulsive forces are applied depending on their relative motion directions. As the angle between the relative distance and velocity decreases, the virtual repulsive force smoothly decreases. However, while the relative distance between the human and the robot remains within the SSM range, the human is still treated as a static obstacle for the calculation of the potential field force, denoted as $F_{b,rep2}(p_b, \hat{p}_m)$:

$$F_{b,rep2}(p_b, \hat{p}_m) = \begin{cases} -r_{b2}\left(\frac{1}{||p_b - \hat{p}_m||} - \frac{1}{d_0}\right)\cdot\frac{p_b - \hat{p}_m}{||p_b - \hat{p}_m||},\ ||p_b - \hat{p}_m|| \le d_0 \\ 0, ||p_b - \hat{p}_m|| > d_0 \end{cases} \tag{18}$$

where d_0 is the maximum calculation distance for static obstacles, calculated based on the robot's stable deceleration performance, as detailed in [46]. The aforementioned repulsive forces can be adjusted by tuning r_{b1} and r_{b2} to ensure continuity at the transition points. These forces are reflected as motion velocities on the mobile platform, and when the combined result of the attractive and repulsive forces exceeds the maximum allowable speed $v_{b\ max1}$ or $v_{b\ max2}$ for the current region, the speed will be capped at that maximum and will no longer increase.

As for the robotic arm, its repulsive force calculation boundary is within the PFL area. Due to the conversion of motion parameters between the task space and the joint space, the repulsive force for the robotic arm is calculated based on the predicted relative position to improve computational efficiency and ensure system stability. Repulsive forces are applied to the positions of the respective joints as follows:

$$F_{a,rep}(p_a, \hat{p}_h) = \begin{cases} -r_a\left(\frac{1}{||p_a - \hat{p}_h||} - \frac{1}{d_1}\right)\cdot\frac{p_a - \hat{p}_h}{||p_a - \hat{p}_h||},\ ||p_a - \hat{p}_h|| \le d_1 \\ 0, ||p_a - \hat{p}_h|| > d_1 \end{cases} \tag{19}$$

where $\hat{p}_h$ represents the position and velocity of the human hand, which can be substituted with the left-hand movement parameters $\hat{p}_{hl}$ or the right-hand movement parameters $\hat{p}_{hr}$. r_a is the repulsive force intensity factor for the robotic arm. For static obstacles, the predicted position $\hat{p}_h$ is replaced with the actual position of the obstacle p_{obs}. This repulsive force acts as a virtual pulling force applied to each joint of the robotic arm. Once synthesized at the end-effector in the task space, it is transformed into the end-effector's linear velocity through the robotic arm's inertia matrix $M(q_a)$:

$$v_{arep} = min\{M^{-1}(q_a)\cdot F_{arep},\ v_{a,max}\} \tag{20}$$

where the $v_{a,max}$ is the maximum allowable safe speed for this area as calculated.

Then, the joint velocities are obtained through the Jacobian matrix $J(q_a)$:

$$\dot{q}_{a^i} = J^{-1}(q_a)\cdot v_{arep} \tag{21}$$

The above equation may encounter singularities in the matrix during the calculation of joint velocities, making it unsolvable. To avoid this issue, a singularity potential field is introduced. This field monitors the motion state of the robotic arm during the obstacle avoidance process. When the arm moves towards a singular configuration, the repulsive force generated by this potential field will prevent the arm from continuing in that direction, allowing the robotic arm to find another stable trajectory before reaching the singular configuration.

4.3. Singularity field

The singularity field monitors the motion state of the robotic arm during the obstacle avoidance process. When the arm moves towards a singular configuration, the repulsive force generated by this potential field will prevent the arm from continuing in that direction, allowing the robotic arm to find another stable trajectory before reaching the singular configuration.

Perform singular value decomposition (SVD) on the Jacobian matrix of the robotic arm:

$$\boldsymbol{J}(p_a) = U\sum V^T \tag{22}$$

where Σ is a diagonal matrix containing the singular values σ_i of the robotic arm's joints. The smaller σ_i is, the closer the robotic arm is to a singular configuration. Based on this, the singularity potential field is designed as follows:

$$U_s(q_{a,i}, \dot{q}_{a,i}) = \begin{cases} -s\dot{q}_{a,i}\left(1 - \frac{\sigma_i}{\sigma_0}\right),\ \sigma_i \le \sigma_0 0,\ \sigma_i > \sigma_0 \end{cases} \tag{23}$$

where σ_0 is an empirical threshold used to determine whether the robotic arm is at risk of a singularity, and it is set based on the arm's configuration. For example, for a six-degree-of-freedom collaborative robotic arm, σ_0 can be set to 0.1. To ensure a smooth change in velocity when the singularity potential field is introduced, the joint velocity $\dot{q}_{a,i}$ is also used as an input for the potential field force calculation, ensuring that the intervention of the singularity potential field does not affect the stability of the robot's motion.

4.4. Stability analysis

In the designed potential field system, the potential field forces represent velocity control rather than traditional dynamic forces. Therefore, we analyze the system's stability based on a velocity control model instead of using traditional dynamic equations. To verify the system's stability, we apply the Lyapunov method and construct a

Lyapunov function based on velocity.

We define a Lyapunov candidate function

$$V(\dot{q}) = \frac{1}{2}\dot{q}^T\dot{q} \tag{24}$$

where $\dot{q}$ is the joint velocity vector of the system, representing the robot's velocity state in task space.

To prove the system's stability, we compute the time derivative of the Lyapunov function $\dot{V}(\dot{q})$:

$$\dot{V}(\dot{q}) = \dot{q}^T\ddot{q} \tag{25}$$

Since the potential field forces are directly represented as velocity control, and assuming the potential field forces are smooth and differentiable, the system's acceleration $\ddot{q}$ can be expressed through a damping control equation:

$$\ddot{q} = -K_d\dot{q} \tag{26}$$

where K_d is a positive definite damping coefficient matrix, indicating that the system decelerates based on the magnitude of the velocity. Substituting this equation into the Lyapunov derivative expression:

$$\dot{V}(\dot{q}) = \dot{q}^T(-K_d\dot{q}) = -\dot{q}^TK_d\dot{q} \tag{27}$$

Since K_d is positive definite, and $\dot{q}^TK_d\dot{q} \geq 0$, we have $\dot{V}(\dot{q}) \leq 0$. This indicates that the system's velocity energy decreases over time, eventually tending to zero, satisfying the conditions for Lyapunov asymptotic stability.

5. Coupled motion module

The platform-arm coupled motion module can only be activated when the robotic arm begins to execute motion. Its purpose is to prevent the robotic arm from entering singular configurations or approaching joint motion limits, which could lead to unstable motion states when subject to multiple potential field constraints. We have designed the following two activation methods to initiate the platform-arm coupled motion, as shown in Fig. 5.

5.1. Activation conditions

First, when the robotic arm approaches a singular configuration due to obstacle avoidance, it may oscillate or stop near the singular position or dangerous area. At this point, the arm's motion is unsafe and inefficient, and it should be promptly removed from this state. During the calculation of the singularity potential field force, if the singular value of certain joints of the robotic arm remains below σ_0 for >1 s, the singularity-avoidance platform-arm coupled motion is triggered. The triggering process for the singularity-avoidance platform-arm coupled motion does not require separate calculations; it only requires adding an abnormal singular value judgment module and a 1-second sliding window to store singular value changes within the singularity potential field calculation, avoiding redundant computations and without increasing the algorithm's complexity. This is the first condition for activating the platform-arm coupled motion.

Second, the aforementioned potential field force for the robotic arm only considers velocity limitations and does not account for joint angle range limits. To enhance the arm's reachability, when the distance between any joint of the arm and its maximum allowable angle is less than 5°, the reachability expansion platform-arm coupled motion is triggered. The monitoring of joint angle limits can typically be achieved by setting soft constraints within the robotic arm's control system to limit its range of motion. When the robotic arm touches the soft constraint boundary, it automatically triggers a warning message. This module only needs to monitor and read this warning message to determine when to trigger the reachability expansion platform-arm coupled motion. This is the second condition for activating the platform-arm coupled motion. As shown in Fig. 5.

5.2. Motion solution

For the singularity-avoidance platform-arm coupled motion, we first extract the left singular vector U_i corresponding to the diagonal matrix element σ_i, where the singular value remains below σ_0 after performing SVD of the Jacobian matrix, as shown in Eq. (22). This vector represents the motion velocity causing the singularity in the robotic arm, with the direction of velocity being in the task space of the end-effector. We then take the first two elements of U_i to form a subvector (u_{xi}, u_{yi}). If both u_{xi} and u_{yi} are not equal to zero, the motion velocity of the mobile platform

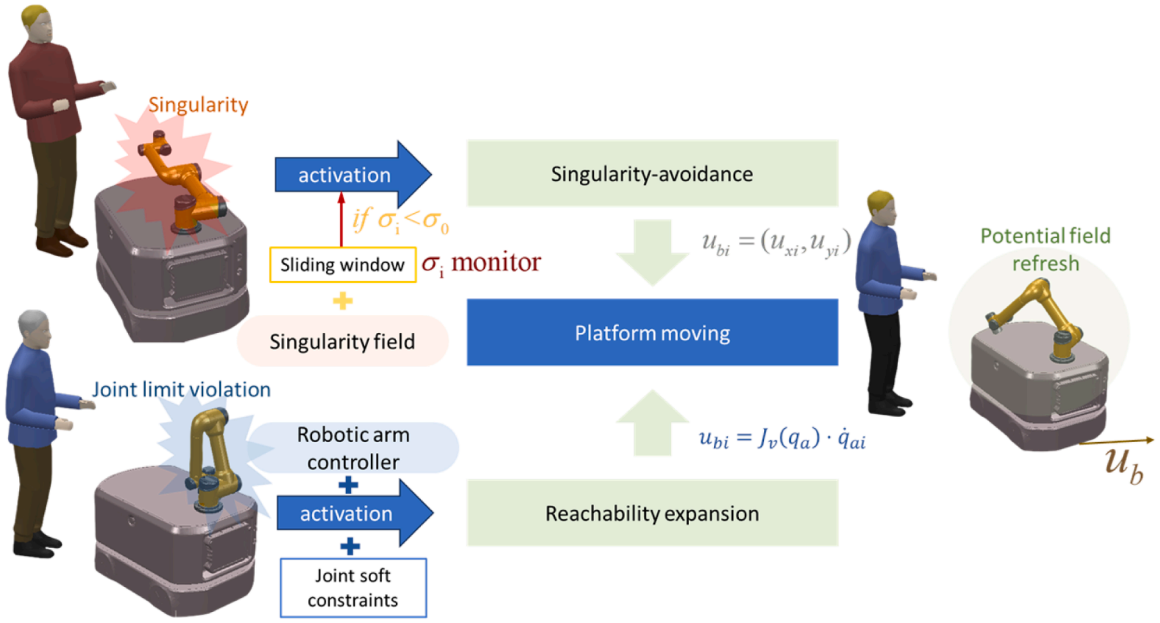


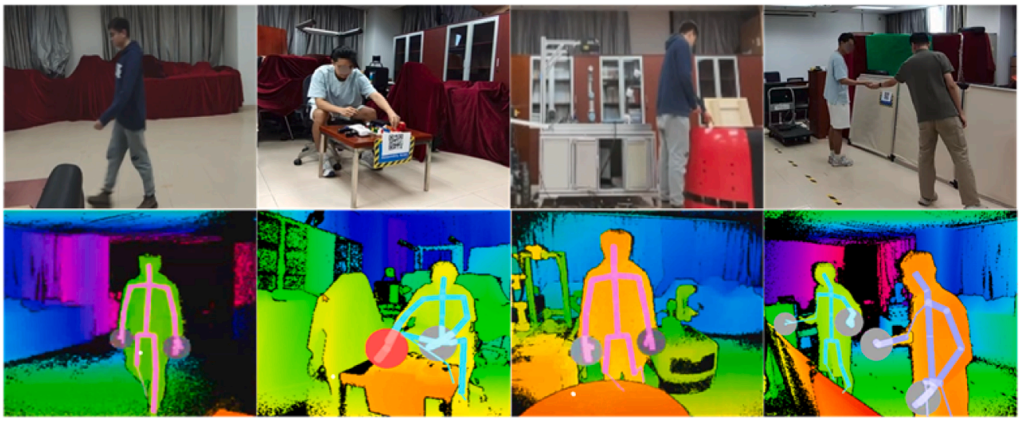
Fig. 5. Flowchart of the platform-arm coupled motion module.

is given by:

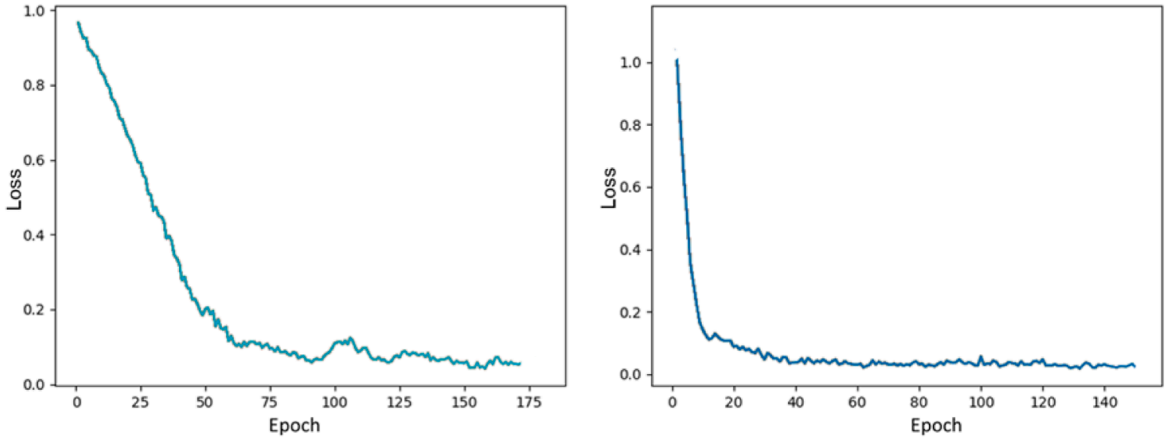
$$u_{bi} = \left[u_{xi}, u_{yi}\right]^T \tag{28}$$

To ensure human-robot safety, the motion distance d_a is limited to 1/10 of a person's stride length d_1. When $u_{xi,}$ and u_{yi} are both zero, meaning that the velocity direction causing the singularity in the task space is along the z-axis, a random perturbation direction is generated. The mobile platform moves a distance d_a of the maximum allowable speed $v_{b\ max1}$. After the movement of the mobile platform, the target pose of the robotic arm in joint space changes, allowing the arm to escape the previous singular configuration and continue smoothly approaching the target pose.

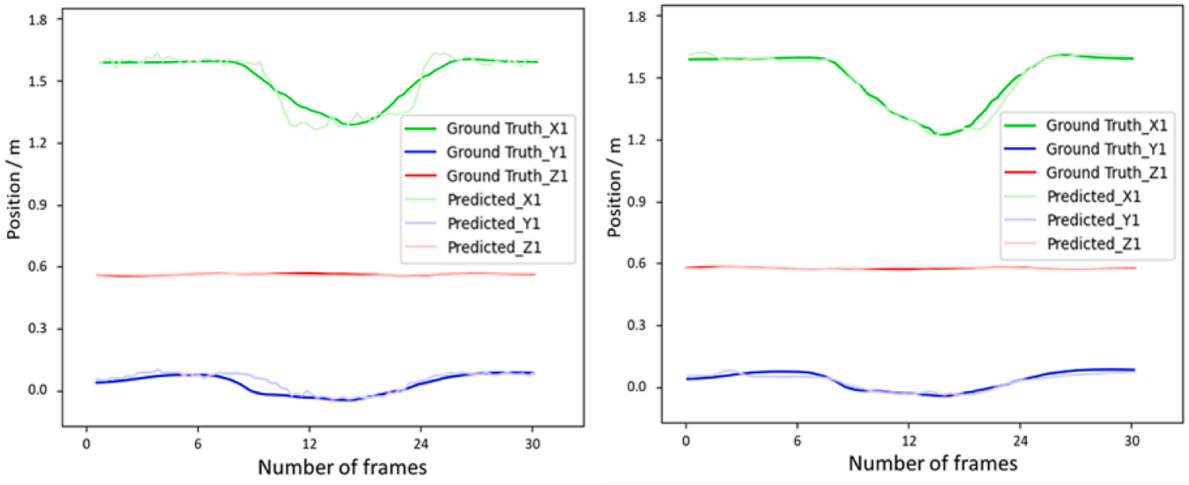
For the reachability expansion platform-arm coupled motion, a motion allocation strategy is designed to transfer part of the robotic arm's movement to the planar motion of the mobile platform. Let the direction of motion of the monitored joint i, which is about to exceed its displacement limit, be denoted as $\dot{q}_i$. The joint velocity vector $\dot{q}_{ai}$ is defined as:



(a) Four task scenarios.



(b) Loss curves during the training process: left is the traditional LSTM, right is the improved LSTM.



(c) Comparison of predicted values and actual values: left is the traditional LSTM, right is the improved LSTM.

Fig. 6. Validation Results of the HRI State Monitoring Module.

$$\dot{q}_{ai} = [0, \cdots \omega_i \cdots, 0]^T \tag{29}$$

Substitute $\dot{q}_{ai}$ into the velocity Jacobian matrix equation:

$$u_{bi} = J_v(q_a) \cdot \dot{q}_{ai} \tag{30}$$

The velocity of the end-effector in task space is obtained. The velocity components in the xy-plane are extracted, and the mobile platform moves a distance d_a along this velocity component. Afterward, the robotic arm updates its target pose and continues its movement towards the target.

6. Experiment and application

In this section, we detail the simulation and real-world experiments conducted to evaluate the effectiveness of the proposed SPF framework. First, the accuracy and complexity of the HRI State Monitoring module were validated using the collected dataset. The rationality of the Potential Field Calculation module was analyzed through numerical simulations, and the stability and efficiency of the improved potential field and Coupled Motion Module in solving robotic trajectories were verified in simulated scenarios. Finally, the feasibility of the entire framework was demonstrated by constructing common industrial scenarios such as inspection, material handling, and assistance in assembly in a real environment.

6.1. Validation of module effectiveness

6.1.1. HRI state monitoring module

The performance of the HRI State Monitoring Module was assessed by validating its ability to accurately predict human actions within HRI spaces. The experiments were performed on a computer equipped with an 11th generation Intel Core i7-1165G7 processor, 16GB RAM, and an NVIDIA GeForce MX450 graphics processor. The operating system is an x64-based 64-bit processor. Python 3.7.16 was used as the programming environment, and PyTorch 1.10.0 (CUDA version) served as the deep learning framework. This configuration has a low deployment cost in industrial scenarios, making it suitable for practical applications in the industry.

As shown in Fig. 6, we collected skeleton data from three individuals performing the same tasks in four scenarios: walking between stations, assembly tasks with significant local movements, processing tasks involving movement across stations, and material transfer tasks involving two people. Each task was executed repeatedly 50 times. To facilitate deployment in industrial settings, the required environmental information inputs for the module were provided in the form of scanned QR codes, which input the semantic vectors of the environment. Data from two individuals were used as the training set for the Improved LSTM, while the data from one individual served as the validation set. To verify the enhancement of training and prediction effectiveness by the introduced scene semantics, a control group was set up using an LSTM network without the Attention mechanism and without processing the environmental semantics. Taking the prediction of the human center hip position during the processing task involving movement across workstations as an example, we output the experimental data for analysis. The loss curves for both models during the training process are shown in Fig. 6(b). It can be observed that the introduction of environmental semantics facilitates faster convergence, thereby improving the efficiency of the training process.

Table 2
The comparison of neural network performance.

Algorithm	Epoch	GPU usage /%	MSE/ m^2	Accuracy/ %	Inference time/ ms
Traditional LSTM	168	69	0.0215	82.6 %	72
Improved LSTM	149	78	0.0081	83.1 %	59

The validation data were used to compare the prediction accuracy of the algorithms before and after the improvements. As shown in Fig. 6(c), the results from the improved algorithm are closer to the actual values. More importantly, the outputs of the improved algorithm are more stable, which contributes to the overall stability of our SPF framework and helps avoid unnecessary disturbances in robot motion planning due to fluctuating prediction data.

As a predictive algorithm applied in industrial scenarios, the time complexity and space complexity of its processing also need to be validated. We compared the prediction performance of the algorithm before and after the improvements. The results are shown in Table 2, where Epoch and GPU usage are calculated during the training process, while MSE, Accuracy, and Inference time are based on the validation process. The analysis reveals that although the improved algorithm has a relatively higher space complexity during the training phase, it remains within acceptable limits for industrial scenarios. Additionally, both the MSE and Accuracy of the prediction results have significantly improved. Furthermore, due to the pruning of multiple irrelevant connections, the improved algorithm exhibits better time complexity during prediction compared to traditional algorithms, which is crucial for deployment in real-world applications.

6.1.2. Potential field calculation module

First, we verify the motion smoothness of the potential field designed in this paper. Fig. 7 shows the attractive potential field force calculated based on the potential field force equation and the corresponding safety constraints, which represent the motion speed of the mobile platform and the robotic arm. From the figure, it can be seen that the speed variation of the robot throughout the motion process is smooth, and the acceleration also changes continuously. Therefore, the process of the robot moving toward the target under the guidance of the attractive force is stable.

Next, we validated the effect of dynamically adjusting the influence radius of the repulsive field for dynamic obstacles. As shown in the left Fig. 7(a), the angle between the relative velocity v_{rel} and the relative distance d_{rel} of the dynamic obstacle and the robot is $<v_{rel}, d_{rel}>$. From the potential field gradient in the left figure and the influence range of the repulsive field in the middle figure, it can be observed that, in this case, the repulsive field only gradually increases its influence radius when there is a potential conflict in relative motion. The robot trajectory in the Fig. 7(b) also demonstrates that this potential field calculation can ensure smooth execution of motion, which proved the effectiveness of the above Eq. (16).

Regarding the impact of obstacles near the target point, we designed a scenario where the target point is within the influence range of the obstacle's repulsive field to verify whether the robot can still smoothly reach the target point, further validating Eqs. (15) and (16). As shown in Fig 8(b), although the dynamic obstacle generates a repulsive potential field near the target position to ensure safety (as seen in the left figure), it does not significantly affect the direction of gradient descent. The robot continues to approach the target with a smoothly varying speed, gradually decelerating (as shown in the middle figure). During the approach, speed limitations were implemented to prevent collisions, resulting in slight deviations in the motion trajectory, yet the robot is still able to approach the target position. This indicates that the potential field design in this paper is reasonable.

For the verification of the robotic arm's potential field, we set up an operator performing routine actions within the arm's reachable space and assigned target poses to the arm. The objective was to verify the success rate and stability of the arm's operation under dynamic interference. Initially, we did not apply the singularity potential field method; instead, the arm moved based solely on the attractive and repulsive forces calculated from Eq. (13) and Eq. (19). The motion trajectory of the arm is shown in Fig. 9(a). Due to the dynamic interference

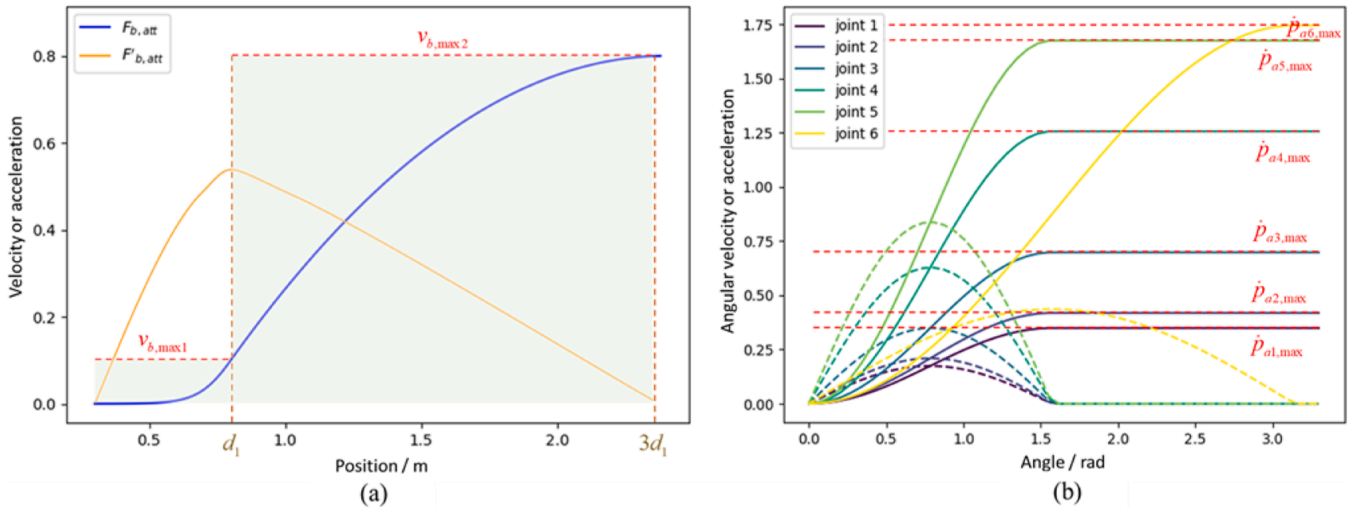
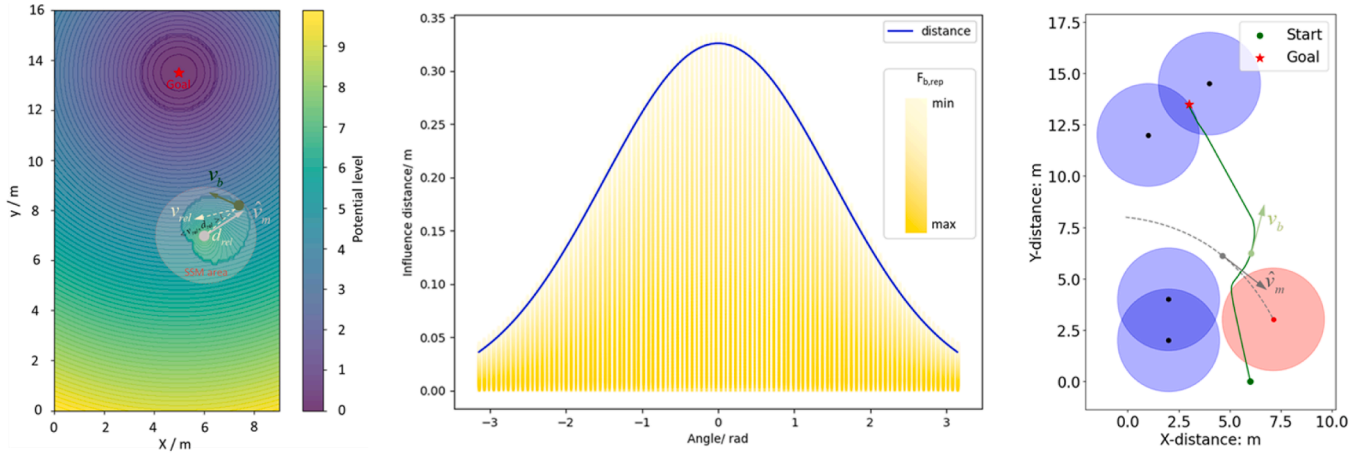
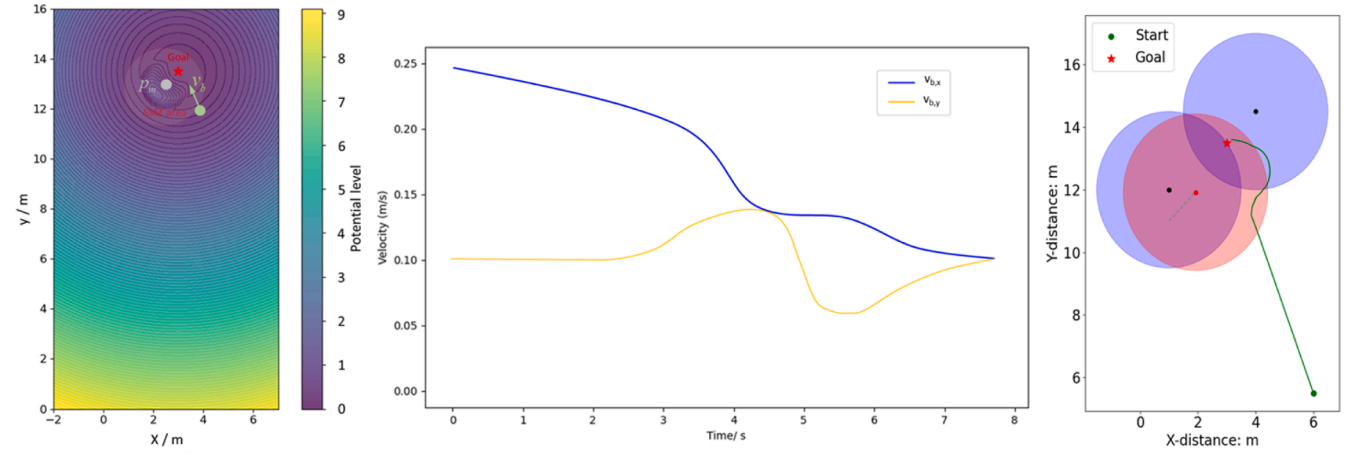


Fig. 7. Attractive potential field force and acceleration variation curves: (a) mobile chassis and (b) joints of the robotic arm.



(a) Verification of potential field handling of dynamic obstacles. Left: potential field gradient map, middle: influence range of repulsive potential field from dynamic obstacles and right: robot motion trajectory..



(b) Verification of potential field handling of obstacles near the target. Left: potential field gradient map, middle: robot approach speed and right: robot motion trajectory.

Fig. 8. Verification of potential field effects for mobile platforms.

from the operator's actions, the arm frequently had to adjust its speed, yet the speed changes remained smooth, indicating that the design of the attractive and repulsive potential fields was reasonable. Additionally, by monitoring the condition number of the arm's Jacobian matrix $\kappa(J(q_a)) = \sigma_{max}/\sigma_{min}$ during the motion, we found that the arm approached singular postures multiple times, posing a risk of instability. Therefore, we introduced the singularity potential field into the trajectory planning process, resulting in a new trajectory as shown in Fig. 9(b). The

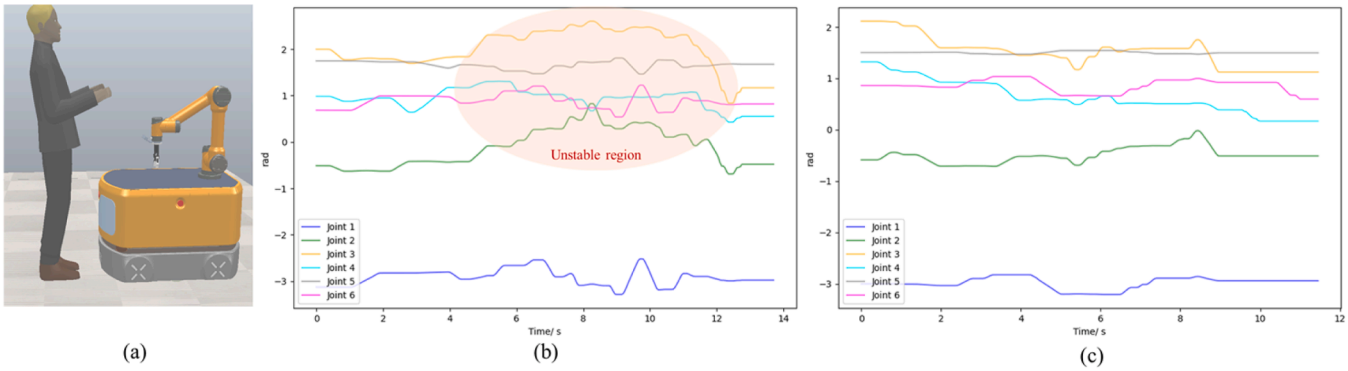


Fig. 9. Robotic arm motion trajectories: (a) experimental scenario, (b) without the introduction of the singularity potential field, and (c) with the introduction of the singularity potential field.

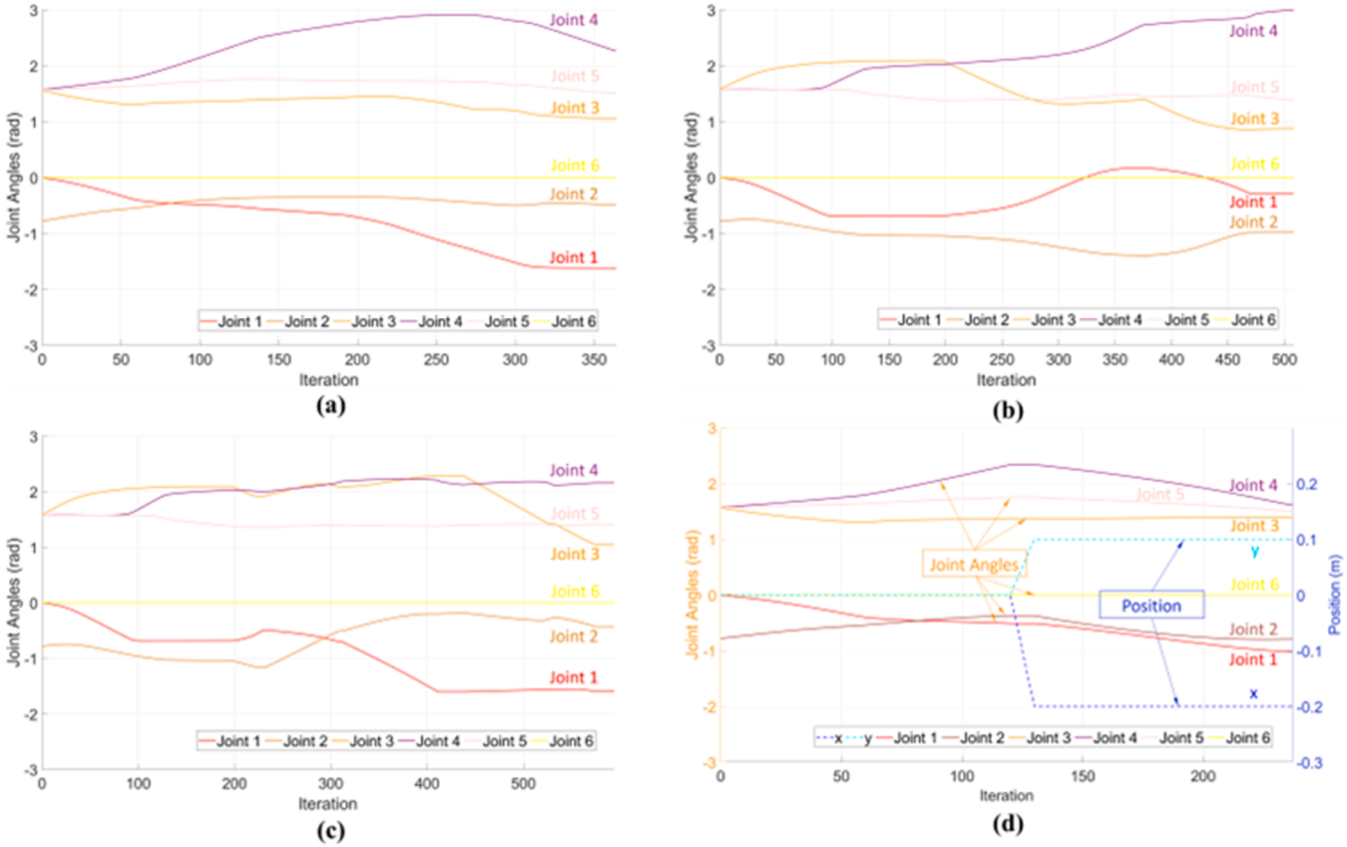


Fig. 10. Robotic arm joint angular displacement: (a) VPF, (b) PAPF, (c) DAPF and (d) improved SPF with platform-arm coupled motion.

condition number remained stable throughout, indicating that the arm successfully avoided singular postures while maintaining smooth motion.

6.1.3. *Coupled motion module*

Here we compared four methods: the velocity potential field (VPF) method facing dynamic obstacles [37], the predictive artificial potential field (PAPF) method incorporating human action prediction [42], the dual artificial potential fields (DAPF) method including singularity calculations [35], and finally, the complete SPF method proposed in this paper including platform-arm coupled movement. Fig. 7 shows the joint angle displacements for these scenarios. Fig. 8 displays the corresponding robot states in the simulation.

As seen in Fig. 10(a) and Fig. 11(a), the VPF method of robotic arm allows the arm to reach the target posture. However, humans are sometimes approached too closely during movement, and accurate avoidance of their movements is difficult. In Fig. 10(b) and Fig. 11(b), the PAPF method is established through the introduction of human action trend prediction, but frequent avoidance maneuvers result in the robotic arm becoming singular and ultimately failing to reach the target posture. In Fig. 10(c) and Fig. 11(c), the DAPF method including singularity calculations ensures that the arm does not fall into singularity during obstacle avoidance, while the process of iteratively searching for the target posture is too lengthy and the operation is not stable. Finally, in Fig. 10(d) and Fig. 11(d), utilizing platform-arm coupled motion, the mobile platform moves, allowing the robotic arm to leave the motion-restricted area near humans and replan its trajectory, safely reaching the target posture.

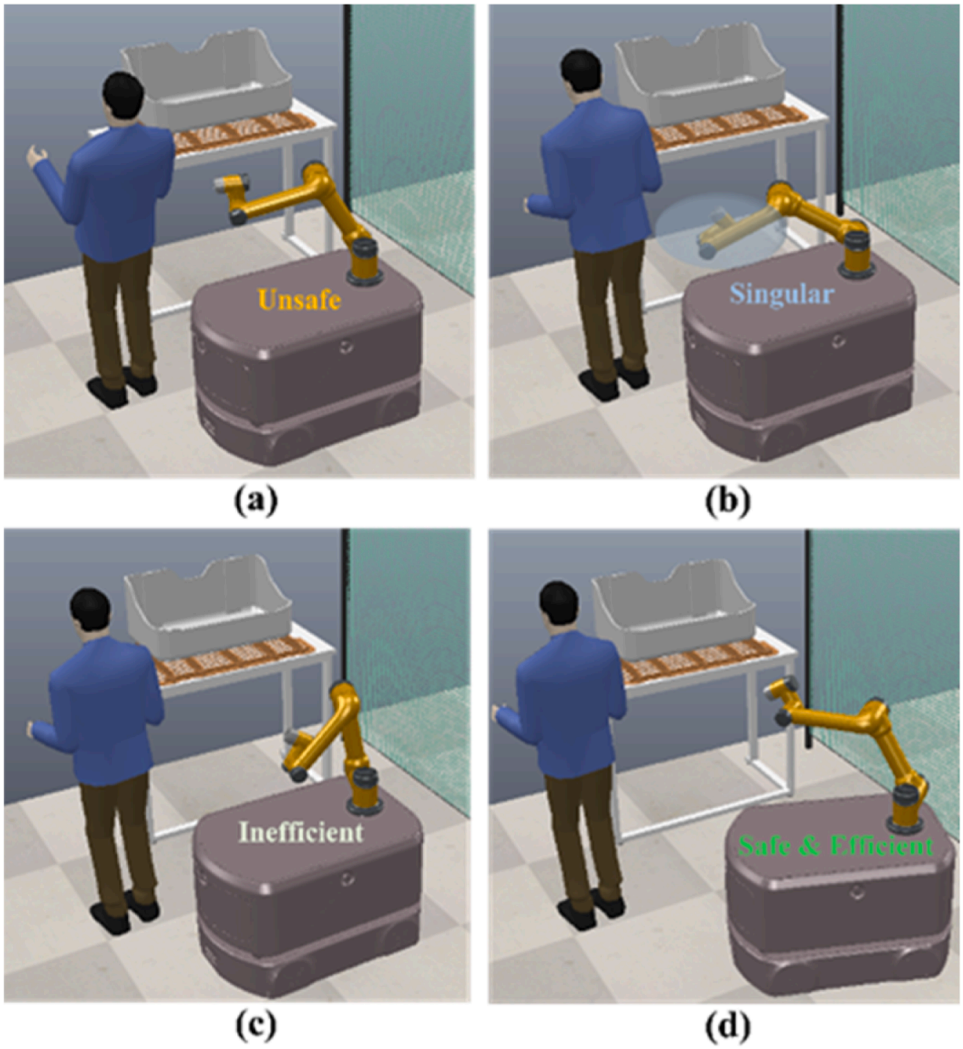


Fig. 11. Robot states: (a) VPF, (b) PAPF, (c) DAPF and (d) improved SPF with platform-arm coupled movement.

Table 3
Comparison of effects of different artificial potential field methods.

Algorithm	Result	Iteration	Time (s)	Maximum Angular Displacement (rad)	Maximum Angular Acceleration (rad/s^2)
VPF	Achieved	365	6.47	1.57	2.13
PAPF	Unachieved	509	15.53	1.57	2.65
DAPF	Achieved	587	21.14	1.57	3.15
Improved SPF	Achieved	232	13.02	0.96	2.12

Table 3 lists a comparison of parameters when using the VPF, PAPF, DAPF and improved SPF method with platform-arm coupled movement.

The data from Table 1 show that the improved SPF reduces the number of iterations in the solution process after introducing platform-arm coupling. However, due to the relative complexity of each iteration and the influence of the mobile platform's movement efficiency, the time to complete operations does not reach the minimum. The algorithm presented in this paper yields significantly smaller maximum angular displacement for the robotic arm joints compared to the other three methods. This indicates that if the robot's computation speed and the mobile platform's operation efficiency are ensured, the operation completion time can be further reduced. The algorithm also results in the smallest maximum angular acceleration of the robotic arm, which can ensure better operational stability.

Fig. 12 illustrates the operational capabilities of the robotic arm under two conditions through the capability map [50]: using only the robotic arm's potential field and introducing the platform-arm coupled SPF. Fig. 9(a) reveals severe operational capability decay in certain areas due to dynamic human obstacles and the arm's own singularity. The method proposed in this paper, as shown in Fig. 9(b), does not exhibit significant degradation in operational capability. This demonstrates its effectiveness in ensuring the safety of the robotic arm's movements and maintaining efficient task execution capabilities.

6.2. Application validation

To validate the advantages of the proposed mobile manipulator SPF trajectory planning method across the entire task execution process, a simulation scenario was created in CoppeliaSim. The scenario simulates a mobile manipulator performing loading and unloading tasks at machine tools within an HRI workshop. Supervisory personnel move back and forth between the machine tools and material platforms, and their

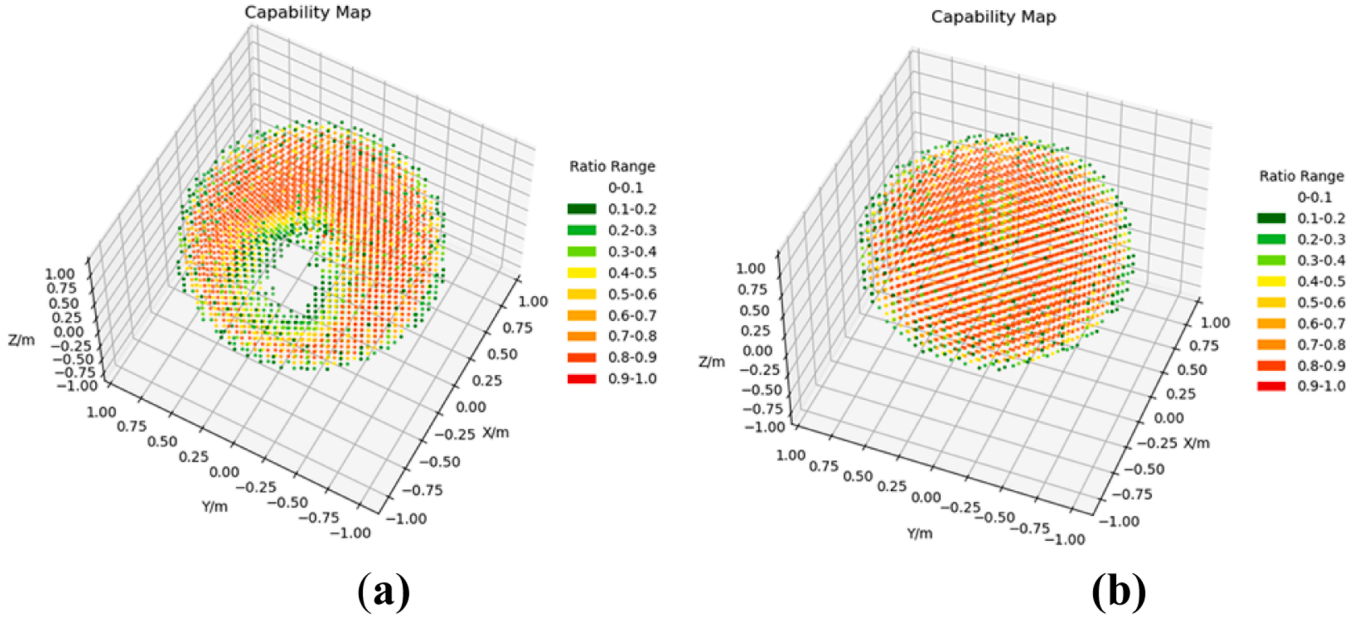


Fig. 12. Robotic arm operational capability map: (a) using only the robotic arm's potential field and (b) introducing the platform-arm coupled motion.

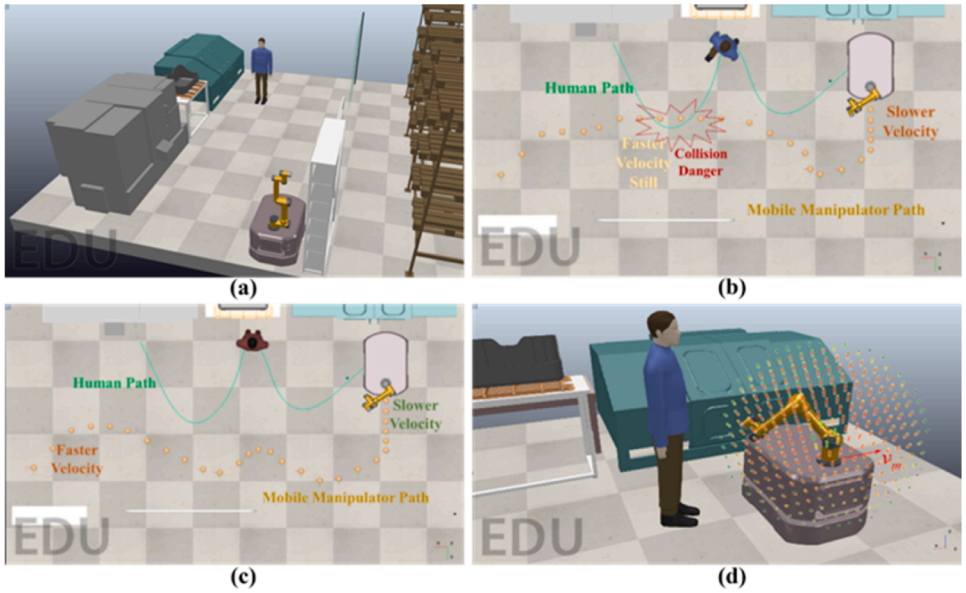


Fig. 13. Simulation scenario: (a) simulation environment scene, (b) the results of the PAPF method, (c) the results of the improved SPF method and (d) the capability map of robotic arm.

real-time movement states are accessible via a remote API. The figure below depicts the motion trajectory of the mobile manipulator and its operational capabilities within the simulation scenario.

Fig. 13 shows the simulation experiment. Among them, Fig. 13(a) is the overall simulation environment scene. We set the human to move between the machine tool and the material site in a semi-circular trajectory. The mobile manipulator moves from the lower right corner of the scene in Fig. 10(a) to the upper left corner, and performs the operation of the manipulator near the human. Fig. 13(b) shows the results of the PAPF method, where the green line is the trajectory of the person and the yellow dots are the positions of the mobile platform. Fig. 13(c) is the results of the improved SPF method. Fig. 13(d) is the capability map of robotic arm after using the improved APF to carry out the coupling motion of the platform-arm.

Comparing Fig. 13(b) and Fig. 13(c), it is evident that by incorporating predicted human action trends as inputs and considering the movement potential of dynamic obstacles over time, the SPF prevents the robot's avoidance trajectory from interfering with human movement in the future. The SPF of the mobile platform ensures current collision-free status while calculating and updating the global motion

Table 4
Comparison of effects of different artificial potential field methods.

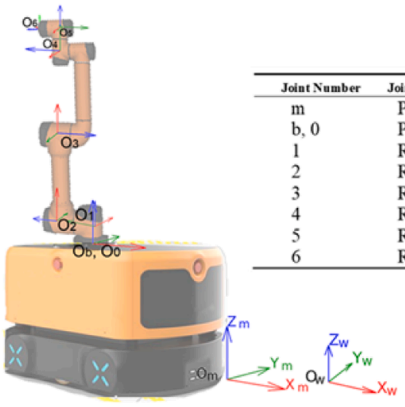
Algorithm	Average time/ s	Success rate/ %	maximum memory usage/ Kb
VPF	26.3	60	6105
PAPF	38.5	80	14,031
DAPF	39.8	60	7934
SPF	34.2	90	21,200

trajectory with minimal safety risk. As the robot approaches the target station, the presence of nearby human motion leads to a reduction in the robot speed to ensure safety while smoothly decelerating toward the target. Moreover, the method minimizes the repulsive force exerted by humans on the robot, ensuring stable arrival at the target station. After reaching the target position, as shown in Fig. 13(d), a high-risk action of personnel approaching the robot was designed in the simulation environment to validate the robot's motion changes in high-risk settings. By calculating the robotic arm's operational capabilities, it was found that approaching personnel affect the arm. Intervention by the mobile platform moves the arm away from the risk area along the plane direction. This restores full operational capability with only the arm's inherent singularity to consider. As a result, task execution efficiency is improved compared to methods that solely constrain the robotic arm.

In the simulation scenario, the entire process of running the mobile manipulator from the starting point of the mobile platform to the target point of the robotic arm was tested. The four aforementioned algorithms were used, each repeated 10 times, and compared based on the average total task completion time, success rate, and maximum memory usage, as shown in Table 4.

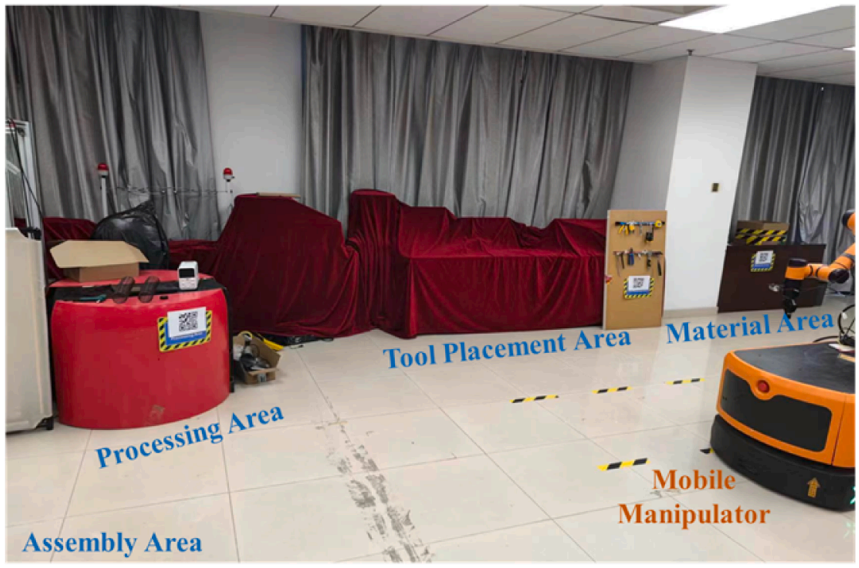
From the analysis of the table, it can be seen that the VPF algorithm only requires direct potential field calculations based on velocity. While it has low memory usage and high execution efficiency, it frequently encounters collisions during task execution, leading to a low success rate, making it unsuitable for industrial deployment. The PAPF algorithm uses a linear prediction method, and its overly conservative motion trajectory results in excessively long task execution times. The DAPF algorithm does not intervene during mobile platform movement, but the singularity potential field significantly compresses the feasible motion space of the robotic arm, resulting in slow solving speeds and frequent unsolvable situations. The only failure of the SPF algorithm occurred due to a communication error between the robot control program and the environment monitoring module, which caused the robot to fail to receive real-time human motion data and stop. Overall, the SPF framework proposed in this paper has advantages in task execution time, balancing both execution efficiency and safe task completion. Although it has relatively higher memory usage, it remains feasible for industrial deployment compared to other more complex non-potential field algorithms.

Furthermore, a simulated industrial scenario was constructed in a real environment. As shown in Fig. 14, The AUBO AMR300 robot was used for movement execution, with motion parameters collected via a Kinect V2 camera. A simulated workshop robot loading and unloading environment was constructed in a real-world setting to validate the method. Kinect V2 cameras were arranged at the experimental site. They monitored personnel actions when the mobile platform arrived and



Joint Number	Joint Type	α_i/°	a_i/mm	θ_i/°	d_i/mm
m	Prismatic	$-\pi/2$	0	$-\pi/2$	700
b, 0	Prismatic	$-\pi/2$	0	$\pi/2$	600
1	Revolute	0	0	θ_1	98.5
2	Revolute	-90	0	θ_2	121.5
3	Revolute	180	408	θ_3	0
4	Revolute	180	376	θ_4	0
5	Revolute	-90	0	θ_5	102.5
6	Revolute	90	0	θ_6	94

(a) Configuration of the mobile manipulator.



(b) Experimental scenario. (c) Experimental equipment.

Fig. 14. The simulated industrial scenario.

during the robotic arm's task execution. Real-time hand and feet coordinate information was then uploaded to the supervisory computer. Human-robot positional relationships were calculated using the method proposed in our improved LSTM was used for action trend prediction, and the robot's SPF was updated based on the coordinate transformation relationship between the camera and the robot. This directed the movement of the robot's mobile platform and robotic arm.

The first simulation scenario setup was as follows: personnel moved from the tool placement area along an arc to assembly area, paused briefly for 3 s, and then moved linearly toward processing area near the robot's target station, simulating operations in the workshop. Meanwhile, after loading from the starting point, the robot executed movement toward the target station near processing area. During this process, dynamic obstacle information was collected via external cameras for real-time avoidance trajectory planning.

To validate the effectiveness of the proposed algorithm, a control experiment using the APF method without HRI prediction was conducted. Since personnel moved from Station A to Station B along an arc trajectory and the nonpredictive method considered this motion to be linear, the robot anticipated that personnel will proceed straight toward a position behind the starting point and execute normal linear movement. After the robot moved linearly for some time, the personnel had already reached the robot's front via the arc trajectory, forcing the robot to halt for avoidance, leading to lower execution efficiency. The SPF algorithm uses monitored historical human motion information to predict human actions as input to the potential field. This allows the robot to accurately avoid humans in advance. As a result, the impact on operational efficiency is minimized. The experimental comparison is shown in Fig. 15.

The second scenario is designed for the robot to transport materials in the turnover area for two people who are passing materials between them. The robot needs to dock between the two individuals. The operation result of the artificial potential field method without calculating the influence range of the dynamic repulsive field is shown in Fig. 16(a). The robot, affected by the two dynamic obstacles, stopped before reaching the target. However, by calculating the dynamic repulsive field, the robot was able to reach the target station at a safe speed, ensuring the successful execution of the task, as shown in Fig. 16(b). Additionally, this experiment verified the ability of the SPF framework proposed in this paper to handle multiple obstacles in the environment. Since the HRI Monitor module is deployed independently on another set of sensor information processing systems, it can successfully solve scenarios where multiple dynamic obstacles are present in the environment.

The third experimental scenario is designed for the robot to perform auxiliary assembly actions near the assembly workstation, as shown in Fig. 17(a). The robot needs to flexibly execute assembly tasks while maintaining a non-contact state with the personnel. Initially, we did not introduce the platform-arm coupled motion module. Due to the influence of the repulsive force from the person's outstretched wrist and the attractive force from the operational task, the robotic arm entered a singular joint configuration, making it impossible to proceed with the next target. The joint motion curve at this time is shown in Fig. 17(b). As the robotic arm approaches the singular posture, the joint velocity changes rapidly, triggering a shutdown, indicating that solely relying on potential field calculations cannot accomplish flexible operations in close proximity to humans. After introducing the platform-arm coupled motion module, when the singularity potential field detects abnormal singular values during the calculation process, the reachability expansion platform-arm coupled motion is triggered. The robotic arm's base coordinates shift a certain distance in the direction causing the singularity, as shown in Fig. 17(c). The robotic arm can then re-calculate its target pose, avoiding the previous singular configuration during obstacle avoidance and approach to the target, and successfully continuing the flexible assembly operation while ensuring safety. The joint angle variation throughout the entire platform-arm coupled motion process is shown in Figure A(d).

The fourth experimental scenario simulates a situation where a person in the processing area requires the robot to deliver tools. Although this scenario involves human-robot interaction after the robot has stopped moving, the robot still needs to perform obstacle avoidance with the person while approaching the target. As shown in Fig. 18(a), during the planning process of the robotic arm towards the collaborative position, without the platform-arm coupled motion module, the robotic arm reaches its joint angle limit due to the influence of only the attractive field, repulsive field, and singularity potential field, as shown in Fig. 18(b). Although the joint angle variation curve is smooth throughout the process, the robotic arm can no longer continue to operate. However, with the introduction of the platform-arm coupled motion, when the robotic arm is about to reach the joint angle limit, the reachability expansion platform-arm coupled motion is triggered, utilizing the robot system's redundant degrees of freedom to expand the operational space of the robotic arm, as shown in Fig. 18(c). This allows the robotic arm to continue performing close-range collaborative tasks while ensuring safety. However, since the robot needs to approach the wrist joint of the person, considered as a dynamic obstacle, the joint angle variations during this process are not as smooth as in other scenarios, as shown in Fig. 18(d). This experiment also provides insights for our future research on improving the safety and efficiency of human-robot physical collaboration.

Additionally, considering that the mobile platform and robotic arm need to monitor and calculate multiple parameters during trajectory planning when computing the potential field, the time complexity and space complexity of the algorithm must be validated to ensure its

Fig. 15. Comparison of robot movement states approaching the target: (a) APF without prediction and (b) improved SPF method.

Fig. 16. The mobile platform approaching the turnover area: (a) fixed repulsive force influence radius and (b) dynamically adjusted repulsive force influence radius.

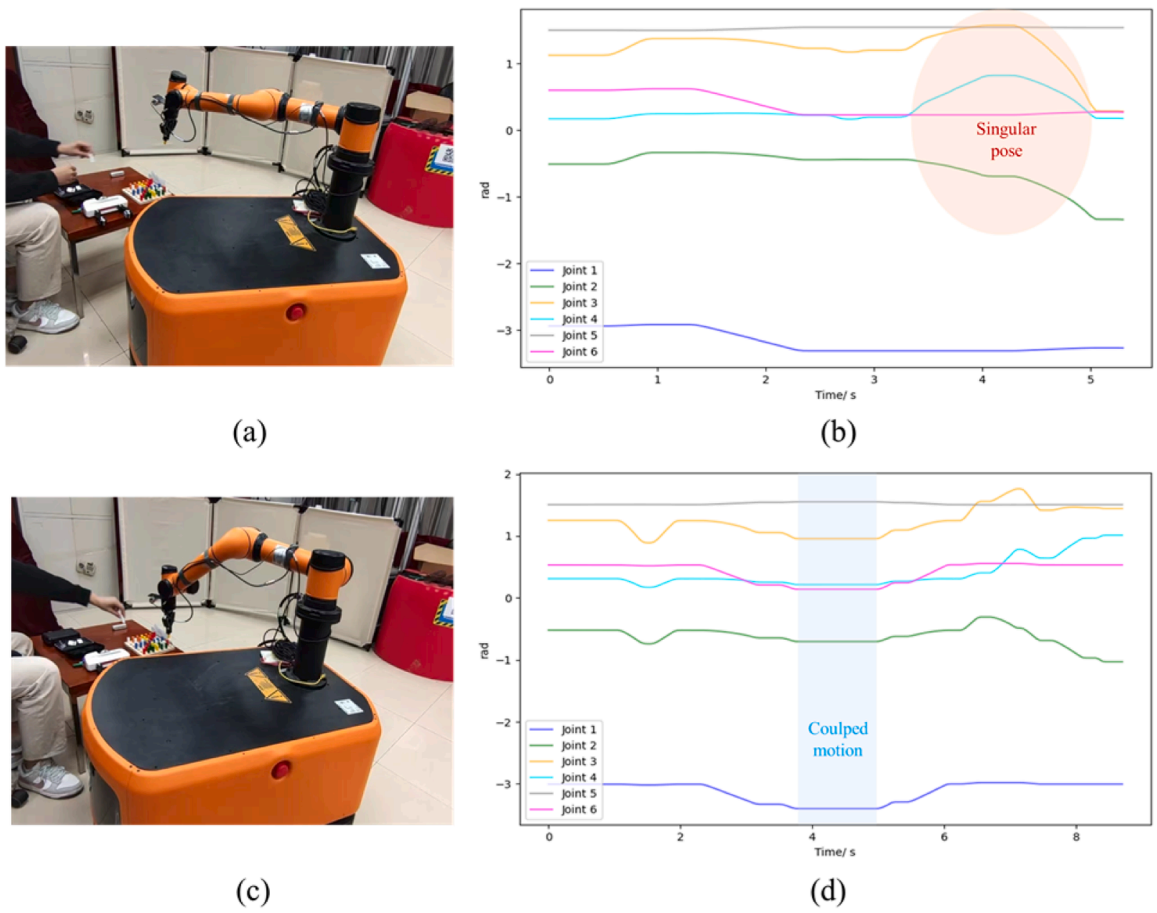


Fig. 17. The mobile manipulator performing auxiliary assembly tasks in the assembly area: (a) and (b) show the operation results and the joint trajectories of the robotic arm without the coupled motion module, (c) and (d) show the results after introducing the coupled motion module.

usability in industrial scenarios. Table 5 presents the computation time and memory utilization of the algorithm during the potential field calculations for the mobile platform and robotic arm. The results are the average values from 10 repeated experiments in the same scenario. The computation time is measured from the moment the robot initiates the next motion step until the system sends the resultant potential field force calculations to the robot's control system. This includes the maximum time for environment data acquisition, potential field computation time, and communication delay. Memory utilization reflects the maximum value during the entire computation process. From the data, it can be seen that the proposed potential field computation method fully meets the requirements for real-time performance and system stability in industrial scenarios. It is important to note that the previously validated HRI state prediction module operates independently in the system, with its results serving as monitoring data inputs to the potential field calculation module. Therefore, parallel processing across modules ensures the real-time performance requirements are met.

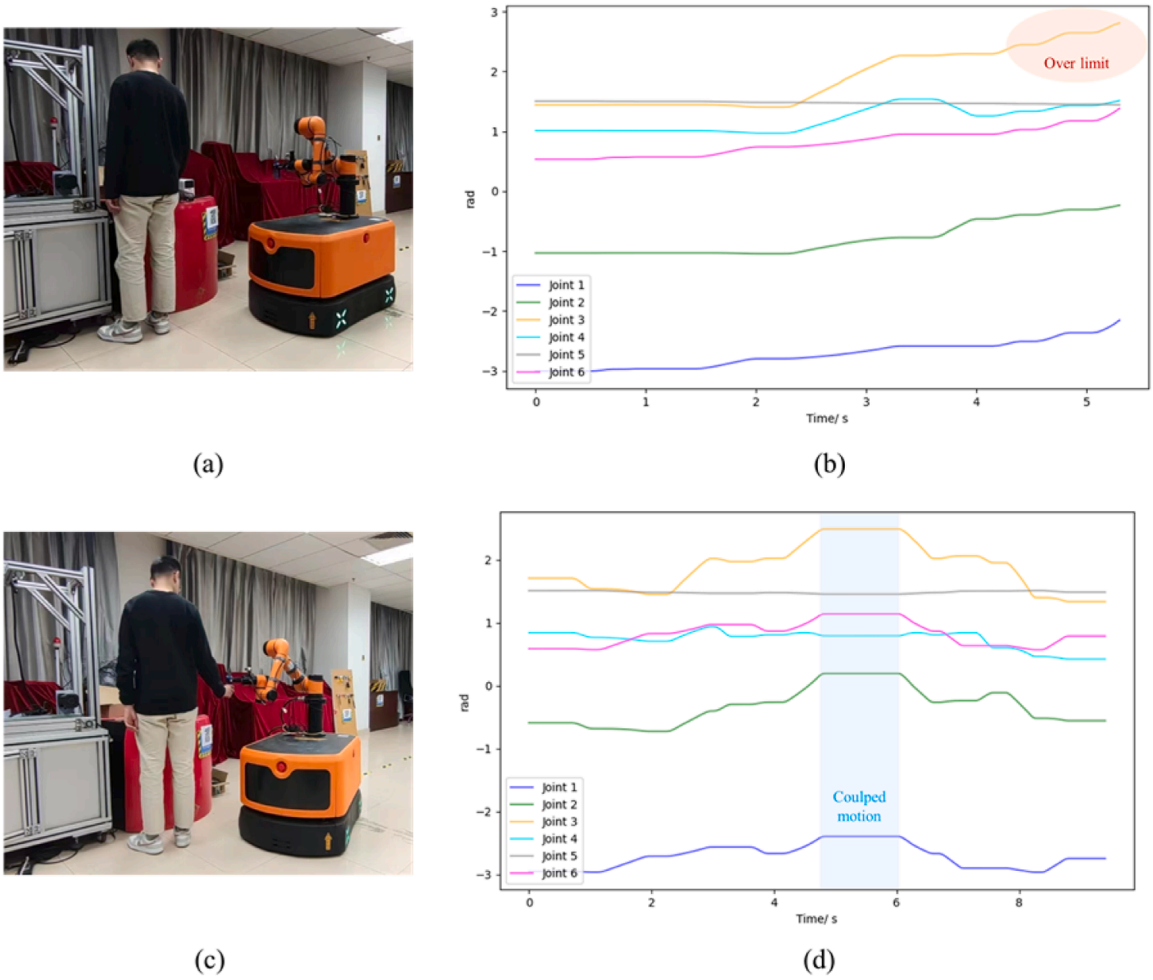


Fig. 18. The mobile manipulator performing tool delivery tasks in the processing area: (a) and (b) show the operation results and the joint trajectories of the robotic arm without the coupled motion module, (c) and (d) show the results after introducing the coupled motion module.

Table 5
The computation time and memory utilization.

Component	Data reading time/ ms	Computing time/ ms	Communication time/ ms	Memory Usage/Mb
Platform	32	18	16	12.3
Arm	26	29	11	21.2

7. Discussion

The Safety Posture Field (SPF) algorithm proposed in this paper demonstrates excellent performance in dynamic human-robot interaction environments, particularly in improving task execution safety and reducing task completion time. Considering the strong correlation between personnel actions and environmental semantics in industrial settings, the algorithm utilizes an improved Attention-LSTM neural network to predict human actions at different workstations, with corresponding network structures and parameter configurations. Although the training process requires significant computational resources, this can be completed prior to actual deployment, thus having little impact on industrial operations. Once trained, the algorithm offers high efficiency and accuracy in predictions. The interactions between the robot's motion, dynamic obstacles, and target positions in human-robot collaboration environments have been analyzed, leading to the design of a dynamically optimized artificial potential field. This avoids excessive obstacle avoidance operations and ensures the robot reaches the target position, while strictly adhering to safety standards for limiting the robot's operating speed. To address potential efficiency limitations and joint space motion constraints caused by singularity potential fields, a platform-arm coupled movement strategy has been proposed, further enhancing the flexibility of mobile manipulators in industrial environments. The algorithm is suitable for deployment in open workshops with clearly defined workstations, such as in tasks like electronic product assembly and inspection, where mobile manipulators can replace traditional assembly lines and conveyor belts. Additionally, the design of the HRI State Monitoring module, Potential Field Calculation module, and Coupled Motion module allows for modular deployment on other devices.

However, the successful execution of this framework depends on the proper configuration of sensing devices. For instance, human motion monitoring currently relies on motion cameras placed at each workstation, with carefully designed angles to prevent obstruction by the robot during operation—an issue that may need to be addressed in industrial deployments. Ongoing research is focused on integrating monitoring devices directly onto the robot. To this end, depth cameras have been added to the end of the robotic arm, and the possibility of embedding data processors into the robot's control system is being explored. Additionally, the algorithm's stability depends on the coordination of the three modules. If one module fails, for example, when the prediction accuracy is insufficient, the fault-tolerant operation of the other modules has not yet been addressed in this work, and further research is needed. Alternatively, the modules could be integrated with more advanced algorithms such as deep reinforcement learning, to

enable validation in more diverse scenarios.

8. Conclusion

The Safe Posture Field framework proposed in this paper achieves a balance between the task execution efficiency and safety of mobile manipulators in dynamic human-robot interaction environments. It designs three modules: HRI state monitoring, potential field calculation, and platform-arm coupled motion. By introducing an improved Attention-LSTM neural network that incorporates environmental information, the model complexity is reduced, enabling more accurate human motion prediction with less response time. Based on human-robot collaboration safety standards and the relative motion trends in dynamic environments, as well as the singularities of the robotic arm, a dynamically optimized artificial potential field is designed to ensure a smoother approach to the target while maintaining safety. When the robotic arm is forced to approach a singular configuration or operate beyond its limits due to excessive potential field constraints, the platform-arm coupled motion strategy is triggered, leveraging the redundant degrees of freedom of the mobile manipulator. This framework ensures the robot's safe collaboration capabilities in dynamic environments, avoiding excessive restrictions on the robot's flexible motion capabilities while further improving the efficiency of trajectory computation. The three modules of this framework can be extended for use in other robotic configurations or integrated with other algorithms in the future. The next step is to continue optimizing the algorithm for tasks involving human-robot physical interaction or highly coordinated human-robot collaboration, to enable mobile manipulators to achieve similar working capabilities as human workers and expand their application in more unstructured industrial scenarios.

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We, the undersigned, declare that:

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All authors listed have contributed significantly to the work presented in this manuscript. The contributions of each author are as follows:

[Yong Tao]: Conceptualization
[Jiahao Wan]: Data Curation and Formal Analysis
[Yian Song]: Methodology
[Xingyu Li]: Review & Editing
[Baicun Wang]: Supervision
[Yiru Wang]: Validation

All authors have reviewed the final version of the manuscript and approve it for submission.

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CRediT authorship contribution statement

Yong Tao: Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Jiahao Wan:** Writing – original draft, Visualization, Validation, Software, Resources, Investigation, Formal analysis, Conceptualization. **Yian Song:** Investigation. **Xingyu Li:** Writing – review & editing. **Baicun Wang:** Writing – review & editing, Methodology. **Tianmiao Wang:** Supervision, Conceptualization. **Yiru Wang:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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